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Distributed Quantum Simulation

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Quantum simulation is a promising pathway toward practical quantum advantage by simulating large-scale quantum systems. In this work, we propose communication-efficient distributed quantum simulation protocols by exploring three quantum simulation algorithms, including the product formula, the truncated Taylor series, and the processing of quantum signals over a quantum network. Our protocols are further shown to be optimal by deriving a lower bound on the quantum communication complexity for distributed quantum simulations with respect to evolution time and the number of distributed quantum processing units. Additionally, our distributed techniques go beyond quantum simulation and are applied to distributed versions of Grover’s algorithms and quantum phase estimation. Our work not only paves the way for achieving a practical quantum advantage by scalable quantum simulation but also enlightens the design of more general distributed architectures across various physical systems for quantum computation.

The pivotal goal of quantum computing is significantly outperforming classical computers in solving certain problems, namely quantum advantage [1–3]. For this objective, a quantum computer with a substantial amount of qubits is required. However, a single quantum processing unit (QPU) faces numerous challenges when scaling up the number of qubits. To fully leverage the potential of quantum power, distributed quantum computing [4–6] that utilizes a network of multiple QPUs, emerges as an essential solution [7–11]. So far, many applications of distributed quantum computing have been proposed, including distributed Shor’s algorithm [12, 13], distributed machine learning [14, 15], etc [16–19]. Besides, quantum simulation, Feynman’s original motivation for proposing quantum computers [20–22], is widely considered one of the most promising applications for achieving practical quantum advantage [23–28]. Yet, a distributed protocol for quantum simulation has not been proposed to date. Our work aims to fill this gap.

Quantum simulation of many-body systems generally involves implementing a dynamical evolution e^{-iHt} for a given Hamiltonian H . There are two types of quantum algorithms for this task. One is the product formula (PF), also known as the Trotter-Suzuki method, which decomposes global dynamics into a product of easily implementable evolutions [29–31]. The other type is referred to as the post-Trotter method, which includes the truncated Taylor series (TS) method [32–34] and quantum signal processing (QSP) method [35–38]. In distributed scenarios, it is natural to ask how to utilize these methods to perform distributed quantum simulation, which involves executing quantum dynamics evolution with multiple QPUs.

In this work, we answer this question by proposing and systematically analyzing three concrete distributed quantum simulation protocols: distributed PF (d-PF), dis-

tributed TS (d-TS), and distributed QSP (d-QSP). These protocols are explored in the context of clustered Hamiltonians [39, 40], which depend on the partitioning of the network. For d-PF, we establish quantum channels to perform the evolution of the Hamiltonian terms involving interactions across the network. For the distributed implementation of post-Trotter protocols (d-TS and d-QSP), the primary challenge lies in the distributed execution of multi-partite entangling quantum operations of interest. We solve this problem by introducing two core subroutines: distributed block encoding and distributed reflection operation with the help of an ancillary node.

Meanwhile, we show the efficiency and the optimality of our distributed quantum simulation protocols. While gate complexity is a major measure of quantum computation complexity, quantum communication complexity typically arises as a more stringent concern in distributed scenarios [41–43]. This is because establishing quantum connectivity in distributed quantum processors is usually much more costly than elementary gates due to the constraints of practical physical systems [44–46]. Regarding this, we prove that our protocols are communication-efficient for simulating generic many-body systems with general quantum networks by establishing a tight lower bound on the quantum communication complexity that depends on the number of parties involved and the evolution time of simulated systems. The d-PF achieves optimal scaling in the number of parties, while the distributed post-Trotter protocols, i.e. d-TS and d-QSP, attain optimality concerning both the number of parties and the evolution time provided that the evolution time is shorter than the system size of a single party.

Additionally, we examine the quantum communication complexity associated with simulating k -local and nearest-neighbor Hamiltonian models. We also demonstrate that our distributed technique can go beyond

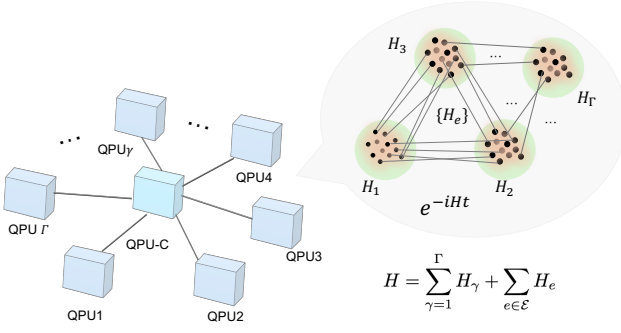


FIG. 1. The star quantum network consists of one control node connecting Γ distributed QPUs. The distributed quantum simulation is to implement e^{-iHt} with a clustered Hamiltonian Eq. (1) induced by the network. H_{γ} is the local term on γ th node (QPU), while H_e across nodes are called interaction terms.

quantum simulation and be applied to Grover's algorithm and quantum phase estimation.

Setup.— In this work, distributed quantum simulation is defined as: given a Γn -qubit Hamiltonian $H = \sum_{l=1}^L H_l$ with $L = \text{poly}(\Gamma n)$ and evolution time t , the goal is to implement the real-time evolution operator e^{-iHt} with Γ distributed QPUs. Each QPU owns n qubits and allows $o(n)$ ancillary qubits for quantum communication or information processing, etc. Without loss of generality, we utilize the star network [47–49] as Fig. 1 to analyze the distributed protocols for quantum simulation [50].

To implement distributed quantum simulation, the partition of qubits for a quantum network has to be considered. Analogous to Refs. [39, 51, 52], in a quantum network, the original Hamiltonian $H = \sum_{l=1}^L H_l$ can be massaged into two parts, i.e.,

$$H = \sum_{\gamma=1}^{\Gamma} H_{\gamma} + \sum_{e \in \mathcal{E}} H_e, \quad (1)$$

which is called the *induced clustered Hamiltonian*. The first summand contains the *induced local* terms $H_{\gamma} := \sum_{l: \text{supp}(H_l) \subseteq v_{\gamma}} H_l$, where v_{γ} is the set of qubits on the γ th node. The second summand contains all remaining terms called *induced interaction* terms that act on multiple nodes. For simplicity, we would omit the words “induced” in the rest of the paper. In the following, we demonstrate how clustered Hamiltonians can aid in various distributed quantum simulation protocols.

Distributed product formula (d-PF).— The product formula (PF) method (also known as the Trotter-Suzuki formula) is deemed the most straightforward way for quantum simulation. Given a local Hamiltonian $H = \sum_{l=1}^L H_l$ and time t , the first-order PF approximates $U = e^{-iHt}$ by the product of the exponentials of the individual operators, i.e. $\tilde{U}(t) := \left(\prod_{l=1}^L e^{-iH_l t/r} \right)^r$ where each exponential can be implemented by elementary quantum

gates efficiently [30]. By the p th-order PF, the segments r to achieve Trotter ϵ can be further improved to $\mathcal{O}(\alpha_{\text{comm},p}^{1/p} t^{1+1/p} / \epsilon^{1/p})$ where $\alpha_{\text{comm},p}$ is the nested commutator norm [52]. In distributed scenarios, with the clustered Hamiltonian as described in Eq. (1), the procedure for our d-PF protocol is as follows. For a single segment, each node (γ) locally performs the evolution $e^{-iH_{\gamma} t/r}$. Then, the evolution of the interaction term $e^{-iH_e t/r}$ is implemented through quantum communication, assisted by the control node as illustrated in Fig. 1, since the support of H_e spans at least two nodes. Consequently, the quantum communication complexity of the d-PF protocol is directly provided by the following theorem.

Theorem 1 (Quantum communication complexity of d-PF). *Given a Γn -qubit clustered Hamiltonian $H = H_0 + \sum_{e=1}^{|\mathcal{E}|} H_e$ with $H_0 := \sum_{\gamma=1}^{\Gamma} H_{\gamma}$ and evolution time $t \in \mathbb{R}^+$, quantum communication complexity of the p th-order distributed product formula (d-PF) to simulate e^{-iHt} within accuracy ϵ is*

$$\mathcal{O}(|\mathcal{E}| \Gamma \tilde{\alpha}_{\text{comm},p}^{1/p} t^{1+1/p} / \epsilon^{1/p}), \quad (2)$$

where $\tilde{\alpha}_{\text{comm},p} := \sum_{e_1, \dots, e_{p+1}=0}^{|\mathcal{E}|} \|[H_{e_1}, \dots, [H_{e_p}, H_{e_{p+1}}]]\|$ is the norm of the nested commutator of terms in H with $H_{e=0} = H_0$, Γ is the number of nodes and $|\mathcal{E}|$ is the number of the interaction terms across multiple nodes.

See the proof of Theorem 1 in Appendix A. Note that the d-PF protocol can be extended to the analog quantum simulation [53–55] with a similar procedure.

Distributed post-Trotter protocols.— In contrast to the Trotter algorithms approximating e^{-iHt} by products, the evolution operator can also be expressed as summation, namely $e^{-iHt} = \sum_k g_k f_k(H)$, where g_k is the coefficient and $f_k(H)$ is the k -th function of H . Two typical examples are the truncated Taylor series (TS) method by the Taylor expansion and the quantum signal processing (QSP) method by the Jacobi-Anger expansion. Generally, these post-Trotter algorithms need block encoding (BE) of Hamiltonians in a larger Hilbert space. The block encoding of an operator H can be regarded as a unitary U with ancillary system a such that $(\langle 0|_a \otimes I_s) U (|0\rangle_a \otimes I_s) = H/\alpha$ where α is a positive number [36]. Typically, the linear combination of unitaries (LCU) is the standard way to implement BE of H [32]. Nonetheless, LCU incurs a probability issue that entails the oblivious amplitude amplification [56], necessitating the reflection operation (RO). Both BE and RO serve as core subroutines in the construction of various algorithms [37]. In a distributed quantum simulation scenario, the distributed implementation of BE and RO should be thoroughly considered. We analyze these implementations based on the distributed architecture depicted in Fig. 2 (a).

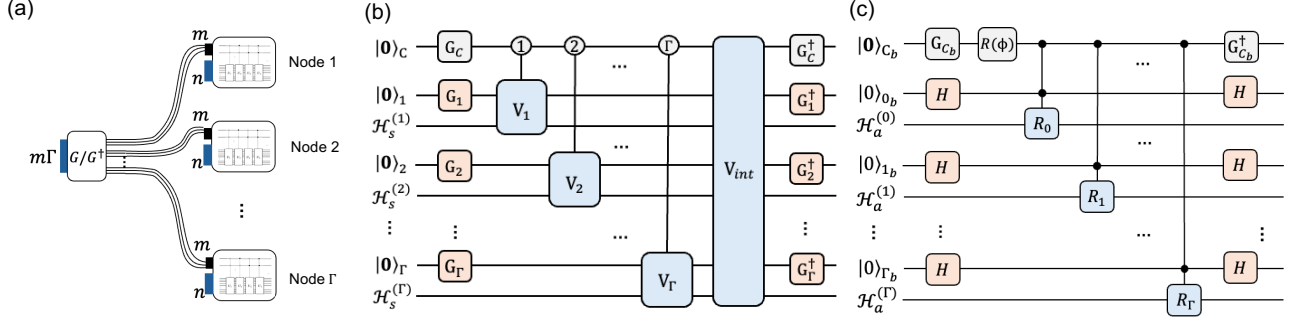


FIG. 2. (a) General distributed architecture for d-BE, d-RO and various algorithms. This distributed architecture utilizes the star network of the control (ancillary) node and Γ local nodes. Each node has n qubits assisted with $m = \mathcal{O}(\log(\Gamma n))$ ancillary qubits, while the control node has $m\Gamma$ qubits for quantum information processing. The general distributed protocols employ the control node to prepare and distribute desired entangled states to each node, which in turn performs global quantum operation and information processing on Γ nodes. Similarly, after conducting the appropriate local operations, each node may return the corresponding auxiliary qubits to the control node for quantum information processing. (b) Quantum circuit for distributed block encoding (d-BE); (c) Quantum circuit for distributed reflection operation (d-RO). In this context, $\mathcal{H}_a$ refers to the Hilbert space corresponding to the auxiliary system in d-BE. For simplicity, we designate the control node as node 0, i.e., $\gamma = 0$.

Distributed Block encoding (d-BE).— We present a distributed quantum circuit to implement d-BE with low quantum communication cost based on a clustered Hamiltonian $H = \sum_{\gamma=1}^{\Gamma} H_{\gamma} + \sum_{e \in \mathcal{E}} H_e$. Recall that H_{γ} is local term support on γ th node, which in principle can be block encoded locally without communication. Contrarily, the interaction terms H_e that cross different nodes require multiple nodes to implement BE together, and their coefficients $\{\|H_e\|\}$ should be shared among the involved nodes. To coherently control all the terms of H for BE (including all H_{γ} and H_e) in a quantum network, the auxiliary system should carry the information of the clustered Hamiltonian. To maintain the nice property that no quantum communication is required for BE of local terms H_{γ} , we are supposed to design a novel d-BE protocol.

Our d-BE circuit entails $\Gamma + 1$ nodes, including one control node and Γ local nodes as in Fig. 2 (b). Specifically, the control node is employed to prepare

$$G_C |0\rangle_C = \frac{1}{\sqrt{\alpha}} \left(\sum_{\gamma=1}^{\Gamma} \sqrt{\|H_{\gamma}\|_1} |\gamma\rangle_C^{\otimes \Gamma} + \sum_{e=1}^{|\mathcal{E}|} \sqrt{\|H_e\|} |e\rangle_C^{\otimes \Gamma} \right), \quad (3)$$

where $\alpha = \|H\|_1 = \sum_{\gamma=1}^{\Gamma} \|H_{\gamma}\|_1 + \sum_{e=1}^{|\mathcal{E}|} \|H_e\|$. Here, $\|\cdot\|$ denotes the operator norm, and $\|\cdot\|_1$ denotes the 1-norm (i.e., the sum of absolute values of Pauli coefficients). Note that $|i\rangle^{\otimes \Gamma}$ is used instead of $|i\rangle$ because, in our distributed protocol, we subsequently have to distribute the prepared auxiliary quantum states to each node. The advantage of using the basis $|i\rangle^{\otimes \Gamma}$ is that it allows the simultaneous transmission of individual qubits to each node, all of which share the same amplitude.

The other Γ ancillary registers are distributed across Γ nodes, each of which is used to perform the BE of the local H_{γ} . Specifically, γ -th ancillary register prepares

$G_{\gamma} |0\rangle_{\gamma} = \frac{1}{\sqrt{\alpha_{\gamma}}} \sum_{l \in v_{\gamma}} \sqrt{\|H_l\|} |l\rangle_{\gamma}$, where $\alpha_{\gamma} = \|H_{\gamma}\|_1$ and v_{γ} denotes the set of qubits on the γ th processor. Similarly, we define the effective select operator as $\text{select}(U) = \sum_{\gamma=1}^{\Gamma} |\gamma\rangle^{\otimes \Gamma} \langle \gamma|_C^{\otimes \Gamma} \otimes V_{\gamma} + V_{int}$ where $V_{\gamma} = \sum_{l \in v_{\gamma}} |l\rangle \langle l|_{\gamma} \otimes \frac{H_l}{\|H_l\|}$ and $V_{int} = \sum_{e=1}^{|\mathcal{E}|} |e\rangle^{\otimes \Gamma} \langle e|_C^{\otimes \Gamma} \otimes \frac{H_e}{\|H_e\|}$ [57] such that the d-BE of H reads

$$\left(\langle 0|_C G_C^{\dagger} \bigotimes_{\gamma=1}^{\Gamma} G_{\gamma}^{\dagger} \right) \text{select}(U) \left(G_C \bigotimes_{\gamma=1}^{\Gamma} G_{\gamma} |0\rangle \right) = \frac{H}{\alpha}, \quad (4)$$

where $|0\rangle = |0\rangle_C \bigotimes_{\gamma=1}^{\Gamma} |0\rangle_{\gamma}$. It can be seen that the d-BE has the same denominator α as the standard BE [36].

The quantum communication cost of d-BE stems from the process of distributing the quantum state $G_C |0\rangle$, which necessitates the transmission of $\Gamma \log(|\mathcal{E}| + \Gamma)$ qubits to Γ nodes. After executing $\text{select}(U)$, the control node needs to perform the $G_C^{\dagger}$ operation, requiring each node to resend the corresponding $\log(|\mathcal{E}| + \Gamma)$ qubits back to the control node. Thus, the overall quantum communication complexity of d-BE is $\mathcal{O}(\Gamma \log(|\mathcal{E}| + \Gamma))$. It should be emphasized that if we employ the original decomposition of Hamiltonian $H = \sum_l^L H_l$ (where L may be significantly larger than $|\mathcal{E}| + \Gamma$), the quantum communication complexity of the d-BE based on it would be $\mathcal{O}(\Gamma \log L)$ which would be unfavorable.

Distributed reflection operation (d-RO). The reflection operation (RO) is another core element of post-Trotter protocols. The RO is defined as $R := I - (1 - e^{-i\phi}) |0\rangle\langle 0|$, where $|0\rangle\langle 0| := |0\rangle\langle 0|_C \otimes \bigotimes_{\gamma=1}^{\Gamma} |0\rangle\langle 0|_{\gamma}$ in our setting (for the all ancillary systems in d-BE). For notation simplicity, we designate the control node as node 0. In this way, the RO can be rewritten as $R = I - (1 - e^{-i\phi}) \bigotimes_{\gamma=0}^{\Gamma} |0\rangle\langle 0|_{\gamma}$. Next, we examine how to perform the distributed reflection operation (denoted d-RO) in a

quantum network. First, we introduce the local reflection operation of the γ -th node as $R_\gamma = 2|\mathbf{0}\rangle\langle\mathbf{0}|_\gamma - I_\gamma$. Then, one may rewrite the operator as

$$R = I - (1 - e^{-i\phi}) \bigotimes_{\gamma=0}^{\Gamma} \left(\frac{R_\gamma + I_\gamma}{2} \right). \quad (5)$$

Since R is unitary and the above decomposition suggests that R can be rewritten as a linear combination of unitary operations, we may introduce new auxiliary systems to implement **d-RO** via the nested LCU technique (see Appendix B 1). In this setting, the auxiliary system $\{C_b, 0_b, 1_b, \dots, \Gamma_b\}$ of the control node and each node for **d-RO** is only 2-dimensional. We present the quantum circuit of **d-RO** in Fig. 2 (c). We prove that the quantum communication complexity of implementing the **d-RO** Eq. (5) is $\mathcal{O}(\Gamma)$ [58]. Remarkably, our scheme can be generalized to the distributed phase operation of any component of interest, i.e. $I - (1 - e^{-i\phi})|j\rangle\langle j|$. Since **RO** also serves as the key operation in Grover's algorithm, we will demonstrate its direct application in the distributed version of Grover's algorithm in the application part.

Communication complexity of post-Trotter protocols.— With the above communication-efficient **d-BE** and **d-RO** designs, the quantum communication complexity of **d-TS** and **d-QSP** are obtained respectively based on the quantum query complexity of these algorithms.

Theorem 2 (Quantum communication complexity of **d-TS**). *Given a Γn -qubit Hamiltonian $H = \sum_{\gamma=1}^{\Gamma} H_\gamma + \sum_{e \in \mathcal{E}} H_e$ and evolution time t , the quantum communication complexity of the distributed TS protocol (**d-TS**) for quantum simulation with accuracy ϵ is*

$$\mathcal{O}(\alpha \Gamma \log(|\mathcal{E}| + \Gamma) K), \quad (6)$$

where Γ is the number of nodes, $|\mathcal{E}|$ is the number of the interaction terms, $K = \frac{\log(\alpha t/\epsilon)}{\log \log(\alpha t/\epsilon)}$ is the truncated order of Taylor series and $\alpha = \|H\|_1$.

Theorem 3 (Quantum communication complexity of **d-QSP**). *Given a Γn -qubit Hamiltonian $H = \sum_{\gamma=1}^{\Gamma} H_\gamma + \sum_{e \in \mathcal{E}} H_e$ and evolution time t , the quantum communication complexity of the distributed QSP protocol (**d-QSP**) for quantum simulation with accuracy ϵ is*

$$\mathcal{O}(\Gamma \log(|\mathcal{E}| + \Gamma) [\alpha t + \log(1/\epsilon)]), \quad (7)$$

where Γ is the number of nodes, $|\mathcal{E}|$ is the number of the interaction edge terms, and $\alpha = \|H\|_1$.

See Appendix B 4 and Appendix B 5 for the proofs of these theorems. It can be seen that the parameters of the interaction graph (i.e., Γ and $|\mathcal{E}|$) crucially determine the quantum communication complexity of these distributed simulation protocols.

Lower bound on quantum communication.— In the context of standard (non-distributed) quantum simulation,

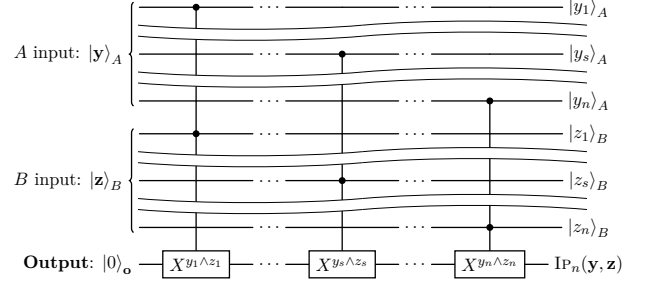


FIG. 3. The construction from an $\text{IP}_n(\mathbf{y}, \mathbf{z})$ circuit (instance) to a Hamiltonian evolution with H_{IP} and initial state $|\mathbf{0}\rangle_{\text{c}} \otimes |\mathbf{y}\rangle_A \otimes |\mathbf{z}\rangle_B \otimes |\mathbf{0}\rangle_{\text{o}}$. Alice (Bob) encodes an n -bit string $\mathbf{y}$ ($\mathbf{z}$) with n -qubit register and the output register consists of one qubit to store the inner product of two bit-strings.

Berry et al. [31] and subsequent works [38, 56, 59] established a lower bound on the quantum complexity, known as the No fast-forwarding theorem. It states that the gate complexity of any generic quantum (dynamics) simulation algorithm scales at least linearly in scaled evolution time $\|H\|t$, which establishes the optimality of the state-of-the-art quantum simulation algorithms. The proof is mainly encoding a hard problem, i.e. evaluating the parity function $\text{PARITY}_N(\mathbf{x})$ of a bit-string $\mathbf{x} \in \{0, 1\}^N$, into an instance of quantum dynamics. In a similar manner, we give a lower bound on the communication complexity of the distributed quantum simulation.

Theorem 4 (Informal, Optimality). *For any positive integer $\tau \leq n$, there exists a sparse Hamiltonian H acting on $2n + \log(n) + 1$ qubits with $\|H\| = \tau$ and a constant evolution time $t = \pi/2$ such that the (bounded-error) quantum communication complexity (QCC_2) of any generic Γ -partite protocol for distributed quantum simulation (**d-QS**) scales at least linearly in scaled evolution time, that is, $\text{QCC}_2(\text{d-QS}) = \Omega(\Gamma \|H\|t)$.*

We start the proof from the bipartite case. The proof is by reducing from the **INNERPRODUCT** problem to the distributed quantum simulation (**d-QS**). The distributed version of **PARITY**, that is, **INNERPRODUCT** is a Boolean function, denoted $\text{IP}_n(\mathbf{x}, \mathbf{y}) : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}$ which evaluates the inner product of two bit-strings. Refs. [41, 60] proved that quantum communication complexity QCC_2 of evaluating IP_n is $\text{QCC}_2(\text{IP}_n) = \Omega(n)$. If two parties could simulate quantum dynamics distributively with $o(\|H\|t)$ quantum communication, then the Boolean function IP_n can be evaluated with $o(n)$ communications which would contradict the known lower bound. Specifically, one can write down a quantum circuit consisting of n Toffoli gates to evaluate the inner product of two bit-strings owned by two parties (see Fig. 3). Correspondingly, a Hamiltonian evolution can be constructed from this circuit such that the evolution can yield the circuit output and the lower bound $\Omega(\|H\|t)$ is obtained.

For a Γ -partite network, Ref. [61] proved that the

lower bound of Γ -partite INNERPRODUCT can be extended to $\Omega(\Gamma n)$. So, our proof can be directly extended to Γ -partite case by replacing the Toffoli gates with Γ -partite controlled-NOT gates and the lower bound becomes $\Omega(\Gamma \|H\|t)$. Since the quantum communication complexity of our post-Trotter protocol is linear with t and Γ , this theorem suggests that our distributed quantum simulation algorithms achieve optimal dependence on evolution time t and the number of partitions Γ provided $\|H\|t \leq n$. We put the complete proof in Appendix F.

Network topology independence.— The diversity of the topological structures in distributed quantum computing architectures is vast [49]. For generality, we argue that the quantum communication complexity of our distributed protocols (except the distributed product formula) is independent of the specific network structure. Intuitively, creating quantum entanglement states in the control node and sharing it with other nodes has the same quantum communication complexity as gradually establishing the entanglement between additional systems at each nearby node (see Appendix E for the proof). Ultimately, networks of arbitrary structure can be virtually effective to be a star network presented in Fig. 1 with the same quantum communication overhead for general distributed protocols.

Applications.— Now we compare the concrete performance of our protocols for simulating common physical Hamiltonians. Consider simulating k -local Hamiltonians $H = \sum_{l_1, \dots, l_k} H_{l_1, \dots, l_k}$, where $H_{l_1, \dots, l_k}$ acts non-trivially on at most k qubits. We show that the quantum communication complexity of the p th-order d-PF is $\mathcal{O}(\min(k, \Gamma)(\Gamma n)^k \|H\|_1 \alpha^{1/p} t^{1+1/p} / \epsilon^{1/p})$ where $\|H\|_1$ is the induced 1-norm [52]. It is proportional to the k th power of the number of qubits in the simulated system, i.e., $(\Gamma n)^k$. In contrast, the post-Trotter protocols (d-TS and d-QSP) have complexity $\mathcal{O}(\alpha \Gamma k \log(\Gamma n) \frac{\log(\alpha t / \epsilon)}{\log \log(\alpha t / \epsilon)})$ and $\mathcal{O}(k \Gamma \log(\Gamma n) [\alpha t + \log(1/\epsilon)])$ respectively which are proportional to $k \log(\Gamma n)$ (see Appendix C).

For a Γn -qubit one-dimensional lattice Hamiltonian with the nearest-neighbor (NN) interaction $H_{\text{NN}} = \sum_{j=1}^{\Gamma n-1} H_{j,j+1}$, the number of interaction terms across terms is proportional to the number of nodes, i.e. $|\mathcal{E}| = \mathcal{O}(\Gamma)$. The quantum communication complexity of the p th-order d-PF, the d-TS, and the d-QSP are $\mathcal{O}(\Gamma^{1+1/p} t^{1+1/p} / \epsilon^{1/p})$, $\mathcal{O}(\alpha \Gamma \log \Gamma \frac{\log(\alpha t / \epsilon)}{\log \log(\alpha t / \epsilon)})$ and $\mathcal{O}(\Gamma \log \Gamma [\alpha t + \log(1/\epsilon)])$, respectively. In this case, the quantum communication complexity for d-PF is more friendly than distributed post-Trotter methods.

Applications beyond quantum simulation.— With our distributed architecture, quantum phase estimation and Grover's algorithm can be easily implemented in quantum networks. For a Γ -partite quantum network and a general Γn -qubit unitary $U =$

$\sum_{j=1}^J \beta_j \bigotimes_{\gamma=1}^{\Gamma} U_j^{(\gamma)}$, the total quantum communication complexity of distributed quantum phase estimation for U is $\mathcal{O}\left(2^{2K} \sqrt{\sum_j |\beta_j| \Gamma \log J}\right)$, where K represents the number of decimals in the phase estimation. Meanwhile, the total quantum communication complexity of distributed Grover's algorithm is $\mathcal{O}(\Gamma \sqrt{N})$, with N denoting the number of items in the database [62].

Discussion.— This work takes a step forward in achieving practical quantum advantage by proposing efficient distributed quantum simulation protocols. On the other hand, previous research has focused on decomposing large-scale quantum computation tasks into smaller partitions using local operations and classical communication only [39, 40, 51, 63–66], typically with exponentially growing complexity. In contrast, our proposed distributed quantum simulation protocols with quantum connectivity may offer an exponential improvement in communication complexity. We also establish the optimality of the quantum communication complexity of our protocol for the number of partitions and time when $t \leq n$. However, achieving optimality for longer times remains challenging.

Meanwhile, in practice, there is potential to further decrease the quantum communication complexity. For example, a more fine-grained partition for a specific Hamiltonian in a certain quantum network may have a lower overhead in quantum communication complexity. Additionally, with the help of variants of quantum simulation algorithms [67–70] and better performance analyses utilizing prior knowledge [71–77], quantum communication complexity for distributed quantum simulation may potentially be further reduced.

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- [1] J. Preskill, [Quantum computing and the entanglement frontier](#) (2012), [arXiv:1203.5813](#).
- [2] F. Arute, K. Arya, R. Babbush, *et al.*, [Nature](#) **574**, 505 (2019), [arXiv:1910.11333](#).
- [3] H.-S. Zhong, H. Wang, Y.-H. Deng, *et al.*, [Science](#) **370**, 1460 (2020), [arXiv:2012.01625](#).
- [4] H. Buhrman and H. Röhrig, in *Math. Found. Comput. Sci. 2003*, Lecture Notes in Computer Science, edited by B. Rovan and P. Vojtáš (Springer, Berlin, Heidelberg, 2003) pp. 1–20.
- [5] R. Beals, S. Brierley, O. Gray, A. Harrow, S. Kutin, N. Linden, D. Shepherd, and M. Stather, [Proc. R. Soc. A](#) **469**, 20120686 (2013), [arXiv:1207.2307](#).
- [6] E. M. Ainley, A. Agrawal, D. Main, P. Drmota, D. P. Nadlinger, B. C. Nichol, R. Srinivas, and G. Araneda, [Multipartite Entanglement for Multi-node Quantum Networks](#) (2024), [arXiv:2408.00149](#).
- [7] H. J. Kimble, [Nature](#) **453**, 1023 (2008).
- [8] M. Elkin, H. Klauck, D. Nanongkai, and G. Pandurangan, in *Proc. 2014 ACM Symp. Princ. Distrib. Comput.*, PODC '14 (Association for Computing Machinery, New York, NY, USA, 2014) pp. 166–175, [arXiv:1207.5211](#).
- [9] S. Wehner, D. Elkouss, and R. Hanson, [Science](#) **362**, eaam9288 (2018).
- [10] F. Le Gall, H. Nishimura, and A. Rosmanis, in *DROPS-IDNv2document104230LIPICsSTACS201949* (Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2019) [arXiv:1810.10838](#).
- [11] M. Caleffi, M. Amoretti, D. Ferrari, J. Illiano, A. Manzolini, and A. S. Cacciapuoti, [Computer Networks](#), 110672 (2024), [arXiv:2212.10609](#).
- [12] A. Yimsiriwattana and S. J. L. Jr, in *Quantum Inf. Comput. II*, Vol. 5436 (SPIE, 2004) pp. 360–372, [arXiv:quant-ph/0403146](#).
- [13] J.-R. Jiang, T.-Y. Wang, W.-H. Huang, and J.-Z. Zhang, in *2023 IEEE 5th Eurasia Conf. IOT Commun. Eng. ECICE* (2023) pp. 479–484.
- [14] H. Tang, B. Li, G. Wang, H. Xu, C. Li, A. Barr, P. Cappelaro, and J. Li, [Phys. Rev. Lett.](#) **130**, 150602 (2023), [arXiv:2209.04888](#).
- [15] C. Li, B. Li, O. Amer, *et al.*, [Phys. Rev. Lett.](#) **133**, 120602 (2024), [arXiv:2310.12893](#).
- [16] L.-Z. Liu, Y.-Z. Zhang, Z.-D. Li, *et al.*, [Nat. Photonics](#) **15**, 137 (2021), [arXiv:2102.11679](#).
- [17] J. Tan, L. Xiao, D. Qiu, L. Luo, and P. Mateus, [Phys. Rev. A](#) **106**, 032417 (2022).
- [18] X. Guo, C. R. Breum, J. Borregaard, S. Izumi, M. V. Larsen, T. Gehring, M. Christandl, J. S. Neergaard-Nielsen, and U. L. Andersen, [Nat. Phys.](#) **16**, 281 (2020), [arXiv:1905.09408](#).
- [19] A. Montanaro and C. Shao, [ACM Trans. Comput. Theory](#) **16**, 1 (2024), [arXiv:2210.01601](#).
- [20] R. P. Feynman, [Int J Theor Phys](#) **21**, 467 (1982).
- [21] R. P. Feynman, [Optics News](#), ON **11**, 11 (1985).
- [22] I. M. Georgescu, S. Ashhab, and F. Nori, [Rev. Mod. Phys.](#) **86**, 153 (2014), [arXiv:1308.6253](#).
- [23] J. I. Cirac and P. Zoller, [Nature Phys](#) **8**, 264 (2012).
- [24] H. Bernien, S. Schwartz, A. Keesling, *et al.*, [Nature](#) **551**, 579 (2017), [arXiv:1707.04344](#).
- [25] C. Monroe, W. C. Campbell, L.-M. Duan, *et al.*, [Rev. Mod. Phys.](#) **93**, 025001 (2021), [arXiv:1912.07845](#).
- [26] E. Altman, K. R. Brown, G. Carleo, *et al.*, [PRX Quantum](#) **2**, 017003 (2021), [arXiv:1912.06938](#).
- [27] Y. Kim, A. Eddins, S. Anand, *et al.*, [Nature](#) **618**, 500 (2023).
- [28] A. J. Daley, I. Bloch, C. Kokail, S. Flannigan, N. Pearson, M. Troyer, and P. Zoller, [Nature](#) **607**, 667 (2022).
- [29] M. Suzuki, [Journal of Mathematical Physics](#) **32**, 400 (1991).
- [30] S. Lloyd, [Science](#) **273**, 1073 (1996).
- [31] D. W. Berry, G. Ahokas, R. Cleve, and B. C. Sanders, [Commun. Math. Phys.](#) **270**, 359 (2007), [arXiv:quant-ph/0508139](#).
- [32] A. M. Childs and N. Wiebe, [QIC](#) **12**, 10.26421/QIC12.11-12 (2012), [arXiv:1202.5822](#).
- [33] D. W. Berry, A. M. Childs, R. Cleve, R. Kothari, and R. D. Somma, [Phys. Rev. Lett.](#) **114**, 090502 (2015), [arXiv:1412.4687](#).
- [34] M. Kieferova, A. Scherer, and D. Berry, [Phys. Rev. A](#) **99**, 042314 (2019), [arXiv:1805.00582](#).
- [35] G. H. Low and I. L. Chuang, [Phys. Rev. Lett.](#) **118**, 010501 (2017), [arXiv:1606.02685](#).
- [36] G. H. Low and I. L. Chuang, [Quantum](#) **3**, 163 (2019), [arXiv:1610.06546](#).
- [37] A. Gilyén, Y. Su, G. H. Low, and N. Wiebe, [Proc. 51st Annu. ACM SIGACT Symp. Theory Comput.](#), 193 (2019), [arXiv:1806.01838](#).
- [38] J. Haah, M. B. Hastings, R. Kothari, and G. H. Low, [SIAM J. Comput.](#), FOCS18 (2021), [arXiv:1801.03922](#).
- [39] T. Peng, A. Harrow, M. Ozols, and X. Wu, [Phys. Rev. Lett.](#) **125**, 150504 (2020), [arXiv:1904.00102](#).
- [40] A. W. Harrow and A. Lowe, [Optimal quantum circuit cuts with application to clustered Hamiltonian simulation](#) (2024), [arXiv:2403.01018](#).
- [41] R. de Wolf, *Quantum Computing and Communication Complexity*, Ph.D. thesis, Universiteit van Amsterdam, Institute for Logic, Language and Computation (2001).
- [42] R. de Wolf, [Theoretical Computer Science Natural Computing](#), **287**, 337 (2002).
- [43] H. Buhrman, R. Cleve, S. Massar, and R. de Wolf, [Rev. Mod. Phys.](#) **82**, 665 (2010), [arXiv:0907.3584](#).
- [44] A. S. Cacciapuoti, M. Caleffi, F. Tafuri, F. S. Cataliotti, S. Gherardini, and G. Bianchi, [IEEE Netw.](#) **34**, 137 (2020), [arXiv:1810.08421](#).
- [45] L. Jiang, J. M. Taylor, A. S. Sørensen, and M. D. Lukin, [Phys. Rev. A](#) **76**, 062323 (2007), [arXiv:0709.4539](#).
- [46] C. Monroe, R. Raussendorf, A. Ruthven, K. R. Brown, P. Maunz, L.-M. Duan, and J. Kim, [Phys. Rev. A](#) **89**, 022317 (2014), [arXiv:1208.0391](#).
- [47] C. Branciard, N. Gisin, and S. Pironio, [Phys. Rev. Lett.](#) **104**, 170401 (2010).
- [48] A. Tavakoli, P. Skrzypczyk, D. Cavalcanti, and A. Acín, [Phys. Rev. A](#) **90**, 062109 (2014), [arXiv:1409.5702](#).
- [49] X. Meng, J. Gao, and S. Havlin, [Phys. Rev. Lett.](#) **126**, 170501 (2021), [arXiv:2103.13985](#).
- [50] At the end of the paper, we will show the quantum communication complexity of general distributed protocols is independent of the specific topology of quantum network.
- [51] A. Lowe, M. Medvidović, A. Hayes, L. J. O’Riordan, T. R. Bromley, J. M. Arrazola, and N. Killoran, [Quantum](#) **7**, 934 (2023), [arXiv:2207.14734](#).
- [52] A. M. Childs, Y. Su, M. C. Tran, N. Wiebe, and S. Zhu, [Phys. Rev. X](#) **11**, 011020 (2021), [arXiv:1912.08854](#).
- [53] E. Farhi and S. Gutmann, [Phys. Rev. A](#) **57**, 2403 (1998), [arXiv:quant-ph/9612026](#).
- [54] T. S. Cubitt, A. Montanaro, and S. Piddock, [Proc. Natl. Acad. Sci.](#) **115**, 9497 (2018), [arXiv:1701.05182](#).
- [55] Z. Liu, D. Devulapalli, D. Hangleiter, Y.-K. Liu, A. J.

- Kollár, A. V. Gorshkov, and A. M. Childs, [Efficiently verifiable quantum advantage on near-term analog quantum simulators](#) (2024), [arXiv:2403.08195](#).
- [56] D. W. Berry, A. M. Childs, and R. Kothari, in *2015 IEEE 56th Annu. Symp. Found. Comput. Sci.* (2015) pp. 792–809, [arXiv:1501.01715](#).
- [57] It should be noted that we have projected the ancillary system onto its encoding basis $\{|0\rangle^{\otimes \Gamma}, |1\rangle^{\otimes \Gamma}, \dots, |i\rangle^{\otimes \Gamma}, \dots\}$.
- [58] See Appendix B 3 for more detail.
- [59] Y. Atia and D. Aharonov, *Nat Commun* **8**, 1572 (2017), [arXiv:1610.09619](#).
- [60] I. Kremer, *Quantum Communication*, Ph.D. thesis, The Hebrew University of Jerusalem (1995).
- [61] F. Le Gall and D. Suruga, in *LATIN 2022: Theoretical Informatics*, Vol. 13568, edited by A. Castañeda and F. Rodríguez-Henríquez (Springer International Publishing, Cham, 2022) pp. 641–657, [arXiv:2210.15402](#).
- [62] See Appendix C for details.
- [63] X. Yuan, J. Sun, J. Liu, Q. Zhao, and Y. Zhou, *Phys. Rev. Lett.* **127**, 040501 (2021), [arXiv:2007.00958](#) [hep-th, physics:quant-ph].
- [64] J. Sun, S. Endo, H. Lin, P. Hayden, V. Vedral, and X. Yuan, *Phys. Rev. Lett.* **129**, 120505 (2022).
- [65] C. Piveteau and D. Sutter, *IEEE Trans. Inf. Theory* **70**, 2734 (2024), [arXiv:2205.00016](#).
- [66] H. Harada, K. Wada, and N. Yamamoto, *PRX Quantum* **5**, 040308 (2024), [arXiv:2303.07340](#).
- [67] Y. Wang and Q. Zhao, *Faster Quantum Algorithms with "Fractional"-Truncated Series* (2024), [arXiv:2402.05595](#).
- [68] A. M. Childs, A. Ostrander, and Y. Su, *Quantum* **3**, 182 (2019), [arXiv:1805.08385](#).
- [69] E. Campbell, *Phys. Rev. Lett.* **123**, 070503 (2019), [arXiv:1811.08017](#).
- [70] Q. Zhao and X. Yuan, *Quantum* **5**, 534 (2021), [arXiv:2103.07988](#).
- [71] B. Şahinoğlu and R. D. Somma, *npj Quantum Inf* **7**, 119 (2021), [arXiv:2006.02660](#).
- [72] W. Gong, S. Zhou, and T. Li, *Quantum* **8**, 1409 (2024), [arXiv:2312.08867](#).
- [73] M. Heyl, P. Hauke, and P. Zoller, *Sci. Adv.* **5**, eaau8342 (2019), [arXiv:1806.11123](#).
- [74] W. Yu, J. Xu, and Q. Zhao, *Observable-Driven Speed-ups in Quantum Simulations* (2024), [arXiv:2407.14497](#).
- [75] Q. Zhao, Y. Zhou, A. F. Shaw, T. Li, and A. M. Childs, *Phys. Rev. Lett.* **127**, 270502 (2021), [arXiv:2111.04773](#).
- [76] C.-F. Chen and F. G. S. L. Brandão, *Commun. Math. Phys.* **405**, 32 (2024), [arXiv:2111.05324](#).
- [77] Q. Zhao, Y. Zhou, and A. M. Childs, *Entanglement accelerates quantum simulation* (2024), [arXiv:2406.02379](#).
- [78] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, 10th ed. (Cambridge University Press, Cambridge ; New York, 2010).
- [79] A. M. Childs and Y. Su, *Phys. Rev. Lett.* **123**, 050503 (2019), [arXiv:1901.00564](#).
- [80] P. W. Shor, *SIAM J. Comput.* **26**, 1484 (1997), [arXiv:quant-ph/9508027](#).
- [81] N. M. P. Neumann, R. Van Houte, and T. Attema, in *Computational Science – ICCS 2020*, Vol. 12142, edited by V. V. Krzhizhanovskaya, G. Závodszy, M. H. Lees, J. J. Dongarra, P. M. A. Sloot, S. Brissos, and J. Teixeira (Springer International Publishing, Cham, 2020) pp. 605–615.
- [82] A. C.-C. Yao, in *Proc. Elev. Annu. ACM Symp. Theory Comput.*, STOC '79 (Association for Computing Machinery, New York, NY, USA, 1979) pp. 209–213.
- [83] M. Christandl, N. Datta, A. Ekert, and A. J. Landahl, *Phys. Rev. Lett.* **92**, 187902 (2004), [arXiv:quant-ph/0309131](#).
- [84] A. Kay, *Int. J. Quantum Inform.* **08**, 641 (2010), [arXiv:0903.4274](#).

Appendix: “Distributed Quantum Simulation”

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Appendix A: Distributed product (Trotter) formula

1. Distributed PF (d-PF) for general Hamiltonians

In this section, we introduce our distributed product formula protocol (denoted **d-PF**) in detail. Given a Γn -qubit Hamiltonian $H = \sum_{l=1}^L H_l$, there is a Γ -partite partition $\mathcal{V}$ of Γn qubits, e.g., $\mathcal{V} = \{v_\gamma : [n\gamma + 1, n(\gamma + 1)]\}$. Based on the specific partition, one may cluster it into two parts, such that

$$H = \sum_{\gamma=1}^{\Gamma} H_\gamma + \sum_{e \in \mathcal{E}} H_e, \quad (\text{A1})$$

where the first summand contains the *local* terms $H_\gamma := \sum_{l: \text{supp}(H_l) \subseteq v_\gamma} H_l$ and the second summand contains all remaining terms called *interaction* terms, which act between two or more two different parties. This type of H is referred to as the (induced) clustered Hamiltonian. Since $\{H_\gamma\}$ represents local Hamiltonian in a quantum network, we can assume that its exponential can be perfectly implemented. For simplicity below, we rewrite Eq. (A1) as

$$H = H_0 + \sum_{e \in \mathcal{E}} H_e = \sum_{e=0}^{|\mathcal{E}|} H_e, \quad (\text{A2})$$

where $H_0 = \sum_{\gamma=1}^{\Gamma} H_\gamma$ and $\sum_{e \in \mathcal{E}} H_e = \sum_{e=1}^{|\mathcal{E}|} H_e$. In the following, we make use of the p th-order product formula [52] to develop a distributed protocol for quantum simulation. The p th-order product formula of clustered Hamiltonian

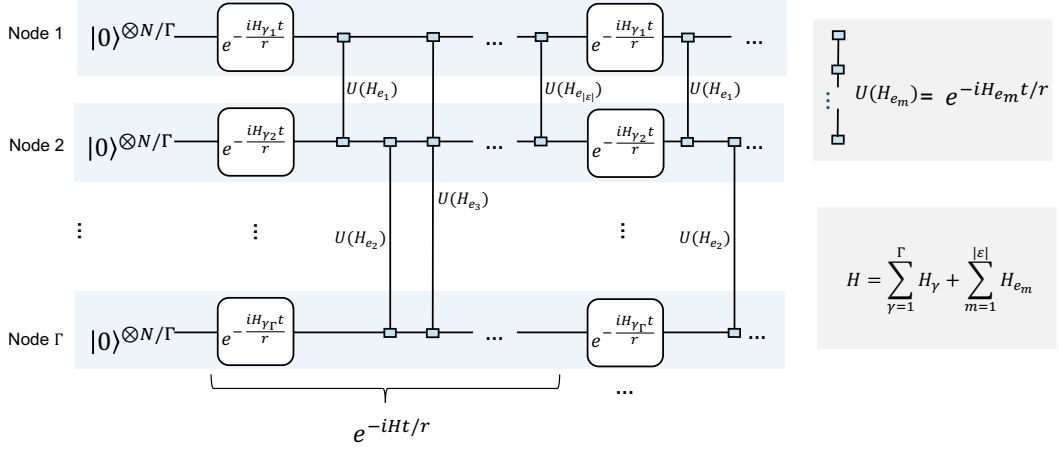


FIG. 4. Quantum circuit of **d-PF** for simulating a Γn -qubit clustered Hamiltonian with a Γ -partite quantum network. Specifically, $U(H_{e_m})$ represents the interaction term, where quantum communication is required for its implementation.

can be written as

$$\tilde{U}_p(t) = \prod_{y=1}^{\Upsilon} \prod_{e=0}^{|\mathcal{E}|} e^{-ita_{(y,e)} H_{\pi_y(e)}}, \quad (\text{A3})$$

where the coefficients $a_{(y,e)}$ are real numbers. The parameter Υ denotes the number of stages of the formula [52]. As a special case, the Suzuki formula with $\Upsilon = 2 \times 5^{p/2-1}$ needs one to alternately reverse the ordering of summands between neighboring stages. Note that Υ and π_y are fixed and the coefficients $a_{y,e}$ are uniformly bounded by 1 in absolute value.

We present the quantum circuit of **d-PF** in Fig. 4 and the procedure of **d-PF** for quantum simulation is as follows:

1. Given the Hamiltonian H , decompose the Hamiltonian according to the graph $H = \sum_{e=0}^{|\mathcal{E}|} H_e$.
2. For a single segment, each node γ performs the local terms $e^{-iH_{\gamma}t/r}$ locally. The interaction terms $e^{-iH_e t/r}$ are implemented through quantum communication since the support of H_e spans at least two nodes.
3. Repeat the above steps according to the formula $\tilde{U}_p(t) = \prod_{y=1}^{\Upsilon} \prod_{e=0}^{|\mathcal{E}|} e^{-ita_{(y,e)} H_{\pi_y(e)}}$.

It can be seen that when H is partitioned, the whole process of **d-PF** is quite straightforward, and quantum communication occurs only in the interaction component $U(H_{e_m})$ across nodes. It is worth noting that the **d-PF** protocol can be extended to the analog quantum simulation [53–55]. In this scenario, during a single time segment, each local node evolves based on its local analog Hamiltonian. At the same time, the interactions between different nodes require the implementation of quantum control, which results in a certain amount of quantum communication cost.

In the following, we will analyze the quantum communication complexity of Hamiltonian simulations with the **d-PF** in detail.

2. Communication complexity of d-PF

Before analyzing the quantum communication complexity of quantum simulations with **d-PF**, we introduce the following Lemma of the p th-order product formula (PF).

Lemma A.1 (Trotter error with commutator scaling [52]). *Let $H = \sum_{l=1}^L H_l$ be a Hermitian operator consisting with L summands, and let $t \geq 0$. Let $\tilde{U}_p(t) = \prod_{y=1}^{\Upsilon} \prod_{l=1}^L e^{-ita_{(y,l)} H_{\pi_y(l)}}$ be a p th-order product formula. Define $\tilde{\alpha}_{\text{comm},p} := \sum_{l_1, \dots, l_{p+1}=1}^L \|[H_{l_1}, [H_{l_2}, \dots, [H_{l_p}, H_{l_{p+1}}]]]\|$. Then, the Trotter error $\|e^{-iHt} - \tilde{U}_p(t)\|$ can be asymptotically*

bounded as $\mathcal{O}(\tilde{\alpha}_{\text{comm},p} t^{p+1})$. To achieve precision ϵ , we need Trotter steps r for $\|e^{-iHt} - \tilde{U}_p^r(t/r)\| \leq \epsilon$, i.e.

$$r = \mathcal{O}\left(\tilde{\alpha}_{\text{comm},p}^{1/p} t^{1+1/p} / \epsilon^{1/p}\right). \quad (\text{A4})$$

Lemma A.1 gives the upper bound of Trotter steps r for p th-order PFs in the general case. Intuitively, for **d-PF**, the quantum communication complexity should be proportional to the Trotter steps r and the number of interaction terms across nodes that require quantum communication.

Theorem A.1 (Quantum communication complexity of **d-PF**, Theorem 1 in maintext). *Given a Γn -qubit clustered Hamiltonian $H = H_0 + \sum_{e=1}^{|\mathcal{E}|} H_e$ with $H_0 := \sum_{\gamma=1}^{\Gamma} H_{\gamma}$ and evolution time $t \in \mathbb{R}^+$, quantum communication complexity of the p th-order distributed product formula (**d-PF**) to simulate e^{-iHt} within accuracy ϵ is*

$$\mathcal{O}(|\mathcal{E}| \Gamma \tilde{\alpha}_{\text{comm},p}^{1/p} t^{1+1/p} / \epsilon^{1/p}), \quad (\text{A5})$$

where $\tilde{\alpha}_{\text{comm},p} := \sum_{e_1, \dots, e_{p+1}=0}^{|\mathcal{E}|} \|[H_{e_1}, \dots, [H_{e_p}, H_{e_{p+1}}]]\|$ is the norm of the nested commutator of terms in H (with $H_{e=0} = H_0$), Γ is the number of nodes and $|\mathcal{E}|$ is the number of the interaction terms across multiple nodes.

Proof of Theorem A.1. As we mentioned before, given an n -qubit Hamiltonian $H = \sum_{l=1}^L H_l$, one may rewrite it in clustered form, i.e. $H = H_0 + \sum_{e \in \mathcal{E}} H_e = \sum_{e=0}^{|\mathcal{E}|} H_e$, where $H_0 = \sum_{\gamma=1}^{\Gamma} H_{\gamma}$ and $|\mathcal{E}|$ is the number of the (induced) interaction terms. In distributed quantum simulation, different parties correspond to quantum processors at different nodes, with local operations within nodes considered ideal, meaning the implementation of $e^{-i\delta t H_0}$ is either perfect or exhibits arbitrarily small error. Therefore, the Trotter step of the p th-order **d-PF** should be

$$r_{\text{d-PF}} = \mathcal{O}\left(\left(\sum_{e_1, \dots, e_{p+1}=0}^{|\mathcal{E}|} \|[H_{e_1}, [H_{e_2}, \dots, [H_{e_p}, H_{e_{p+1}}]]]\| \right)^{1/p} t^{1+1/p} / \epsilon^{1/p}\right), \quad (\text{A6})$$

which is the same as the Trotter number for simulating the clustered Hamiltonian with circuit cutting (see Appendix

I in [52]). That is, in this partition, $\tilde{\alpha}_{\text{comm},p} := \sum_{e_1, \dots, e_{p+1}=0}^{|\mathcal{E}|} \|[H_{e_1}, [H_{e_2}, \dots, [H_{e_p}, H_{e_{p+1}}]]]\|$.

For 2-local Hamiltonians, the quantum communication complexity of each segment is $\mathcal{O}(|\mathcal{E}|)$ since each induced interaction term $e^{-i\delta t H_{e \in \mathcal{E}}}$ requires $\mathcal{O}(1)$ quantum communication to be realized. For example, $H_{e_m} = \|H_{e_m}\| \sigma_{x_i} \otimes \sigma_{z_j}$, where $i \in v_i$, $j \in v_j$ and $i \neq j$. The implementation of $e^{-i\delta t \|H_{e_m}\| \sigma_{x_i} \otimes \sigma_{z_j}}$ requires $\mathcal{O}(1)$ qubits to be teleported or transferred from v_i to v_j , and vice versa.

On the other hand, let us consider the interaction term of the more general k -fold tensor product of Pauli operators and $k \geq \Gamma$, e.g., $H_{e_m} = \|H_{e_m}\| \bigotimes_{j=1}^{\Gamma} (\bigotimes_{i \in v_j} \sigma_{q_i})$, where $1 \leq m \leq |\mathcal{E}|$ and $q_i = \{0, x, y, z\}$ represents the four Pauli operators. This type of Hamiltonian has non-trivial effects on all or nearly all parts of a large system. Unlike the two-local Hamiltonian, which can be realized directly by moderating interactions, the evolution of such a Hamiltonian typically may require at least one auxiliary qubit. A standard approach to perform $e^{-i\delta t \|H_{e_m}\| \bigotimes_{j=1}^{\Gamma} (\bigotimes_{i \in v_j} \sigma_{q_i})}$ is presented in [78], in which one ancillary qubit needs to be entangled with qubits of interaction. Without loss of generality, we assume $H_{e_m} = \|H_{e_m}\| \bigotimes_{j=1}^{\Gamma} (\bigotimes_{i \in v_j} \sigma_{z_i})$. Specifically,

$$e^{-i\delta t \|H_{e_m}\| \bigotimes_{j=1}^{\Gamma} (\bigotimes_{i \in v_j} \sigma_{z_i})} = \left(\langle 0|_a \otimes I_S \left(\prod_{j=1}^{\Gamma} \prod_{i \in v_j} \text{CNOT}_{i,a} \right)^{\dagger} \right) \left(e^{-i\delta t \|H_{e_m}\| \sigma_z \otimes I} \right) \left(\left(\prod_{j=1}^{\Gamma} \prod_{i \in v_j} \text{CNOT}_{i,a} \right) |0\rangle_a \otimes I_S \right), \quad (\text{A7})$$

where $\text{CNOT}_{i,a}$ is the CNOT gate of i th qubit in v_j (control qubit) and ancillary qubit (target qubit). I_S represents the identity operator on the whole distributed quantum system.

The cost of quantum communication required to implement $\prod_{i \in v_j} \text{CNOT}_{i,a}$ is $\mathcal{O}(1)$ since the auxiliary qubit can be transmitted to the v_j th node by performing quantum teleportation only once. Hence, the quantum communication complexity involved in executing $\prod_{j=1}^{\Gamma} \prod_{i \in v_j} \text{CNOT}_{i,a}$ is $\mathcal{O}(\Gamma)$. In other words, the quantum communication cost of $e^{-i\delta t \|H_{e_m}\| \bigotimes_{j=1}^{\Gamma} (\bigotimes_{i \in v_j} \sigma_{z_i})}$ is $\mathcal{O}(\Gamma)$. Similarly, by performing an appropriate single-qubit rotation, the same procedure

can be employed to perform single $e^{-i\delta t \|H_{e_m}\| \bigotimes_{j=1}^{\Gamma} (\bigotimes_{i \in v_j} \sigma_{q_i})}$ with quantum communication complexity $\mathcal{O}(\Gamma)$. Considering that there are $|\mathcal{E}|$ induce interaction terms (H_e) in each segment, the quantum communication complexity of each segment is $\mathcal{O}(|\mathcal{E}|\Gamma)$ in general. Synthesizing the Trotter number of Trotter step of the clustered Hamiltonian, we obtain an upper bound on the quantum communication complexity of d-PF as $\mathcal{O}(|\mathcal{E}|\Gamma \tilde{\alpha}_{\text{comm},p}^{1/p} t^{1+1/p}/\epsilon^{1/p})$. $\square$

Appendix B: Distributed post-Trotter protocols

Besides the product formula (Trotter method), the other promising class of quantum simulation algorithms is called post-Trotter methods. The post-Trotter methods, including the Truncated Taylor series method (TS) and quantum signal processing (QSP) mainly expand the exponential function as a linear combination of functions of H . That is, generally, Hamiltonian simulation using the post-Trotter methods needs block encoding (BE) of matrix H in the quantum circuits or quantum states. Typically, the linear combination of unitaries (LCU) technique [32] is the standard way to implement BE of H [36]. However, LCU presents a probabilistic issue that requires the incorporation of oblivious amplitude amplification [56], necessitating another core operation, the reflection operation (RO). Below we will introduce distributed versions of LCU, BE, RO, TS, and QSP step by step.

1. Distributed linear combination of unitaries (d-LCU)

In the main text, while there is no specific introduction to distributed LCU (d-LCU), it serves as the foundation for the implementation of distributed quantum algorithms, such as distributed Block encoding (d-BE) and distributed reflection operation (d-RO). This is because both d-BE and d-RO fundamentally rely on the nested LCU technique, which is crucial for their operation and effectiveness in distributed quantum computing tasks. Therefore, it is essential to recognize the significance of d-LCU as a building block in the development of distributed quantum algorithms, e.g., distributed quantum phase estimation and distributed Grover's search.

Here we briefly review the LCU method presented in [33]. Suppose our goal is to implement an n -qubit unitary operation V on a quantum computer. As prior knowledge, V can be decomposed into $V = \sum_{j=1}^J \beta_j U_j$, where U_j is unitary and β_j is corresponding coefficient. There is a unitary G transforming the ancillary state $|\mathbf{0}\rangle$ to

$$G|\mathbf{0}\rangle = \frac{1}{\sqrt{s}} \sum_{j=1}^J \sqrt{\beta_j} |\mathbf{j}\rangle, \quad (\text{B1})$$

where s is the normalized parameter, $s = \sum_{j=1}^J \beta_j$. The select U is defined as $\text{select}(U) := \sum_j |\mathbf{j}\rangle\langle\mathbf{j}| \otimes U_j$, where the control register $|\mathbf{j}\rangle\langle\mathbf{j}|$ acts on the ancillary system $\mathcal{H}_{\text{anc}}$. Constructing an operator as $W := (G^\dagger \otimes I) \text{select}(U) (G \otimes I)$. When the input state is $|\mathbf{0}\rangle \otimes |\psi\rangle$, acting W on the global system, one has

$$W|\mathbf{0}\rangle \otimes |\psi\rangle = \frac{1}{s} (|\mathbf{0}\rangle \otimes V|\psi\rangle) + \frac{\sqrt{s^2 - 1}}{s} |\phi\rangle, \quad (\text{B2})$$

where $|\phi\rangle$ is orthogonal to the subspace $\text{span}\{|\mathbf{0}\rangle \otimes \mathcal{H}_{\text{sys}}\}$. After projecting to $|\mathbf{0}\rangle$ of the ancillary system, one may obtain the desired operation on the target system with probability $\frac{1}{s}$. However, the probability of failure in some tasks should not be underestimated, so we need to amplify the probability of success. When $s = 2$, we can boost the probability through the oblivious amplitude amplification protocol,

$$-WRW^\dagger RW|\mathbf{0}\rangle \otimes |\psi\rangle = |\mathbf{0}\rangle \otimes V|\psi\rangle, \quad (\text{B3})$$

where $R := (I - 2|\mathbf{0}\rangle\langle\mathbf{0}|) \otimes I$ is the reflection operation acting on $\mathcal{H}_{\text{anc}} \otimes \mathcal{H}_{\text{sys}}$. For the case $s > 2$, one may decompose V into $V = \prod_{j=1} V^{(j)}$, in which each $V^{(j)} = \sum_{i=1} \alpha_i^{(j)} U_i^{(j)}$ and $s^{(j)} = \sum_{i=1} \alpha_i^{(j)} = 2$.

a. The d-LCU protocol

Here we extend the LCU technique to a quantum network. Consider a system with a control node consisting of $m\Gamma$ qubits and Γ local nodes, where each local node is made up of n working qubits and m auxiliary qubits. The distributed LCU protocol (denote d-LCU) is to implement the $n\Gamma$ -qubits operation V in the quantum network.

As prior knowledge, V may be decomposed as $V = \sum_{j=1}^J \beta_j \bigotimes_{\gamma=1}^{\Gamma} U_j^{(\gamma)}$ where $U_j^{(\gamma)}$ is a unitary of γ -th node and $m := \lceil \log(J) \rceil$. Next, the control node prepares the following entangled state

$$G|\mathbf{0}\rangle := \frac{1}{\sqrt{s}} \sum_{j=1}^J \sqrt{\beta_j} |\mathbf{j}\rangle^{(1)} \otimes |\mathbf{j}\rangle^{(2)} \otimes \dots \otimes |\mathbf{j}\rangle^{(\Gamma)} \equiv \frac{1}{\sqrt{s}} \sum_{j=1}^J \sqrt{\beta_j} \bigotimes_{\gamma=1}^{\Gamma} |\mathbf{j}\rangle^{(\gamma)}, \quad (\text{B4})$$

where the normalization factor $s = \sum_{j=1}^J \beta_j$, each $|\mathbf{j}\rangle$ is encoded by m qubits and $|\mathbf{j}\rangle^{(\gamma)}$ indicates that these m qubits will be transmitted to the γ -th node (using quantum teleportation).

Upon receiving the ancillary state Eq. (B4) from the control nodes, each node executes a locally controlled unitary operation $\sum_{j_\gamma} |j_\gamma\rangle\langle j_\gamma| \otimes U_{j_\gamma}^{(\gamma)}$, and the global operation is $\bigotimes_{\gamma=1}^{\Gamma} \left(\sum_{j_\gamma} |j_\gamma\rangle\langle j_\gamma| \otimes U_{j_\gamma}^{(\gamma)} \right)$. Note that the encoding basis for the ancillary states prepared in the control node is $\{|0\rangle^{\otimes \Gamma}, |1\rangle^{\otimes \Gamma}, \dots, |J\rangle^{\otimes \Gamma}\}$. Thus the effective $\text{select}(U)$ in the quantum network is given as $\Pi \bigotimes_{\gamma=1}^{\Gamma} \left(\sum_{j_\gamma} |j_\gamma\rangle\langle j_\gamma| \otimes U_{j_\gamma}^{(\gamma)} \right) \Pi$ where $\Pi = \bigotimes_{\gamma=1}^{\Gamma} |j\rangle^{(\gamma)} \langle j|^{(\gamma)} = \sum_j |j\rangle^{\otimes \Gamma} \langle j|^{\otimes \Gamma}$, i.e.

$$\text{select}(U) := \sum_j |\mathbf{j}\rangle^{\otimes \Gamma} \langle \mathbf{j}|^{\otimes \Gamma} \bigotimes_{\gamma=1}^{\Gamma} U_j^{(\gamma)} \quad (\text{B5})$$

Similar to the non-distributed case, we construct an operator as $W := (G^\dagger \otimes I) \text{select}(U) (G \otimes I)$. When the initial state is $|\mathbf{0}\rangle \otimes \left(\bigotimes_{\gamma=1}^{\Gamma} |\psi_\gamma\rangle \right)$, performing W on the whole system, we have

$$W|\mathbf{0}\rangle \otimes \left(\bigotimes_{\gamma=1}^{\Gamma} |\psi_\gamma\rangle \right) = \frac{1}{s} \left(|\mathbf{0}\rangle \otimes V \bigotimes_{\gamma=1}^{\Gamma} |\psi_\gamma\rangle \right) + \frac{\sqrt{s^2 - 1}}{s} |\phi\rangle, \quad (\text{B6})$$

where $|\phi\rangle$ is orthogonal to the subspace $\text{span} \{ |\mathbf{0}\rangle \otimes \mathcal{H}_{sys}^{\otimes \Gamma} \}$. After projecting to $|\mathbf{0}\rangle$ of the ancillary systems, we may obtain the desired operation on the target system with probability $\frac{1}{s}$. However, the probability of failure in some tasks can be significant, hence, we need to amplify the probability of success. When $s = 2$, we can boost the probability through the oblivious amplitude amplification protocol,

$$-WRW^\dagger RW|\mathbf{0}\rangle \otimes \left(\bigotimes_{\gamma=1}^{\Gamma} |\psi_j\rangle \right) = |\mathbf{0}\rangle \otimes V \left(\bigotimes_{\gamma=1}^{\Gamma} |\psi_j\rangle \right), \quad (\text{B7})$$

where $R := (I - 2|\mathbf{0}\rangle\langle\mathbf{0}|) \otimes I$ is the reflection acting on $\mathcal{H}_{anc}^{\otimes \Gamma} \otimes \mathcal{H}_{sys}^{\otimes \Gamma}$.

The procedure of **d-LCU** is summarized as follows:

1. The control node uses an $m\Gamma$ -qubit processor to prepare entangled states $\frac{1}{\sqrt{s}} \sum_{j=1}^J \sqrt{\beta_j} \bigotimes_{\gamma=1}^{\Gamma} |\mathbf{j}\rangle^{(\gamma)}$.
2. The control node dispatches the prepared entangled states to each of the Γ nodes via Γ quantum channels, with each node accepting m qubits.
3. Upon receiving the quantum states from the control node, the Γ nodes store them in their local auxiliary quantum systems, each consisting of m qubits. At this stage, each local node contains m qubits of information regarding the entangled state distributed by the control node, as well as an n -qubit local quantum state.
4. Each local auxiliary m -qubit system starts to interact with the local n -qubit system to carry out the corresponding unitary operation $U_j^{(\gamma)}$ ($\gamma \in [\Gamma]$).
5. The information of the respective m qubits from the Γ nodes after the local operations, is transmitted back to the control system, where the $G^\dagger$ operation is performed and the $m\Gamma$ qubits are measured to achieve the desired operation V on the $n\Gamma$ qubits (if amplitude amplification is performed, instead of measuring qubits, R should be performed). Since this process is probabilistic, the use of amplitude amplification, achieved by repeating the previous steps, can help to execute the target operation on the Γn qubits in a nearly deterministic manner.

Note that any operation can in principle be written in the form of LCU, thus universal quantum gates can also be implemented in the **d-LCU** protocol, leading to universal quantum computation in quantum networks.

Communication complexity of d-LCU. In the previously mentioned d-LCU protocol, both quantum and classical communications occur during the process for distribution of entanglement from the control node to the Γ nodes, and the process for the transmission of the auxiliary state from the Γ nodes back to the control node. Two rounds of quantum teleportation involving $m\Gamma$ qubits are necessary to execute a single W operation. Consequently, a W operation demands the transmission of $2m\Gamma$ qubits of information and $4m\Gamma$ classical bits. It is important to note that the number of ancillary qubits, m , is dependent on the size of L , i.e., $m = \lceil \log(J) \rceil$.

- In the case of $s = 2$, three W operations are required to achieve the desired $n\Gamma$ -qubit operation, which corresponds to a consumption of $6m\Gamma$ qubits and $12m\Gamma$ classical bits for quantum and classical communication, respectively.
- For the case $s > 2$, one may decompose V into $V = \sum_{j=1} \beta_j^{(k)} \bigotimes_1^\Gamma U_j^{(k,1)}$, in which each $V^{(k)} = \sum_{i=1} \beta_i^{(k)} \bigotimes_{\gamma=1}^\Gamma U_i^{(k,\gamma)}$ and $s^{(k)} = \sum_{i=1} \beta_i^{(k)} = 2$. In this case, the communication complexity is $6r\Gamma \lceil \log J \rceil$ qubits and $12r\Gamma \lceil \log J \rceil$ bits for quantum and classical communication, respectively. In general, by $\mathcal{O}(\sqrt{s})$ queries, one may perform V with quantum and classical communication complexity $\mathcal{O}(\sqrt{s}\Gamma \log J)$.

In the following section, we only focus on quantum communication complexity and we will show that d-LCU is a gadget that assists in implementing d-BE and d-R0.

2. Distributed block encoding (d-BE)

Before we introduce distributed TS and QSP protocols, it is necessary to design a construction for the block encoding of H within a quantum network [33, 36]. As previously mentioned in the d-PF section, given a Γn -qubit Hamiltonian $H = \sum_l H_l$, there is an interaction graph of clustered Hamiltonian $H = \sum_{\gamma=1}^\Gamma H_\gamma + \sum_{e \in \mathcal{E}} H_e$ with Γ nodes and $|\mathcal{E}|$ interaction terms. In the following sections, we will demonstrate that compared to the decomposition $H = \sum_l H_l$ (where H_l are the k -fold tensor product of Pauli operators), the $\sum_{\gamma=1}^\Gamma H_\gamma + \sum_{e \in \mathcal{E}} H_e$ may give a significantly low quantum communication complexity for d-BE in general cases.

Our task is to implement the Γn -qubit block encoding of H within a Γ -node quantum network, each node containing $n + \log(n)$ qubits where $\log(n)$ qubits are ancillary qubits. Recall the standard block encoding of $H = \sum_{l=1}^L H_l$ [36]

$$\langle \mathbf{0} | G^\dagger \otimes I \left(\sum_{l=1}^L |\mathbf{l}\rangle \langle \mathbf{l}| \otimes \frac{H_l}{\|H_l\|} \right) G | \mathbf{0} \rangle \otimes I = \frac{H}{\alpha}, \quad (\text{B8})$$

where $G | \mathbf{0} \rangle = \sum \frac{1}{\sqrt{\alpha}} \sqrt{\|H_l\|} |l\rangle$ and $\alpha = \|H\|_1$ ($\|\cdot\|$ and $\|\cdot\|_1$ are the spectral norm and 1-norm, respectively). Let's now delve into how to implement a distributed block encoding within a quantum network.

To reduce quantum communication, we adopt the clustered Hamiltonian representation for block encoding. This approach eliminates the need to encode Γ local Hamiltonians using quantum communication. As shown in Appendix B 2 (a), we present the nested block encoding for clustered Hamiltonian $H = \sum_{\gamma=1}^\Gamma H_\gamma + \sum_{e \in \mathcal{E}} H_e$ in a quantum network. The d-BE can be interpreted as either a linear combination of BEs or a nested linear combination of unitaries.

The procedure for d-BE is as follows. To begin, the control node prepares

$$G_C | \mathbf{0} \rangle_C = \frac{1}{\sqrt{\alpha}} \left(\sum_{\gamma=1}^\Gamma \sqrt{\|H_\gamma\|_1} |\gamma\rangle_C^{\otimes \Gamma} + \sum_{e=1}^{|\mathcal{E}|} \sqrt{\|H_e\|} |e\rangle_C^{\otimes \Gamma} \right), \quad (\text{B9})$$

where $\alpha = \|H\|_1 = \sum_{\gamma=1}^\Gamma \|H_\gamma\|_1 + \sum_{e=1}^{|\mathcal{E}|} \|H_e\|$. Note that $|i\rangle^{\otimes \Gamma}$ is used instead of $|i\rangle$ because, in our distributed protocol, we subsequently have to distribute the prepared auxiliary quantum states to each node. The advantage of using the basis $|i\rangle^{\otimes \Gamma}$ is that it allows the simultaneous transmission of individual qubits to each node, all of which share the same quantum amplitude $\{\sqrt{\|H_\gamma\|_1}, \sqrt{\|H_e\|}\}$.

The remaining Γ ancillary registers are distributed across Γ nodes, with each register used to execute the block-encoding of the local H_γ . Specifically, γ -th ancillary register prepares $G_\gamma | \mathbf{0} \rangle_\gamma = \frac{1}{\sqrt{\alpha_\gamma}} \sum_{l \in v_\gamma} \sqrt{\|H_l\|} |l\rangle_\gamma$ where $\alpha_\gamma = \|H_\gamma\|_1$ and v_γ represents the set of qubits on the γ th processor. From the standpoint of the standard BE of LCU, the distributed G operation for the entire ancillary system which includes control nodes and auxiliary systems for each local node is expressed as $G = G_C \otimes \bigotimes_{\gamma=1}^\Gamma G_\gamma$. In a manner similar to d-LCU, we define the effective **select**(U) :=

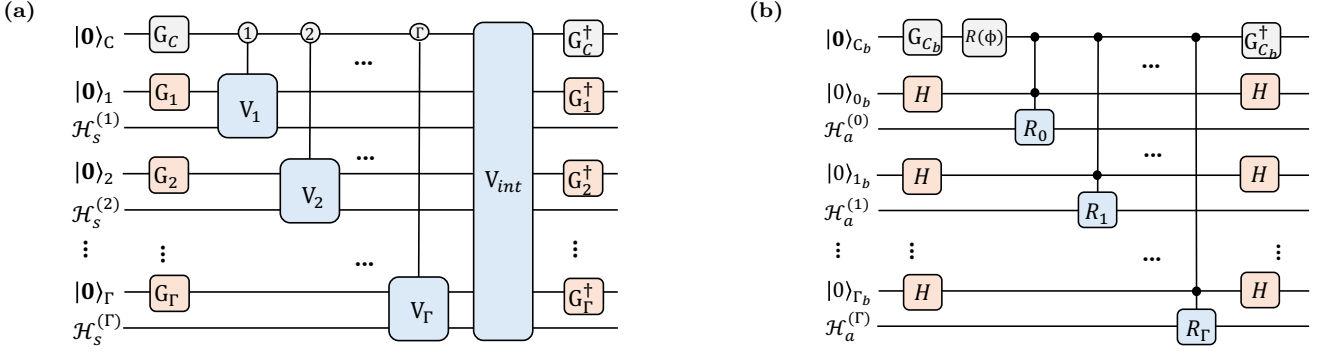


FIG. 5. (a) Distributed block encoding (d-BE) of $H = \sum_{\gamma=1}^{\Gamma} H_{\gamma} + \sum_{e \in \mathcal{E}} H_e$ using d-LCU in a quantum network with Γ nodes and a control node. The control node prepares $G_C |0\rangle$ as Eq. (B9). The implementation of V_{int} (for $\{H_e\}$) requires that the qubits of the control node are needed to be transmitted to the entire network. (b) Distributed reflection operation (dro). In a quantum (star) network, the distributed reflection operation (d-RO) can act on the qubits of the Γ nodes and the central control node. For simplicity, we designate the control node as node 0, i.e., $\gamma = 0$. The newly introduced ancillary system C_b of the control node is then used to facilitate the implementation of d-RO. Without loss generality, $|i\rangle_{C_b}$ can be defined as $|i\rangle_{C_b} = |i\rangle \otimes |i\rangle^{\otimes \Gamma}$, where each qubit in $|i\rangle^{\otimes \Gamma}$ should be individually sent to each of the Γ nodes.

$\sum_{\gamma=1}^{\Gamma} |\gamma\rangle^{\otimes \Gamma} \langle \gamma|_C^{\otimes \Gamma} \otimes V_{\gamma} + V_{\text{int}}$ which is explicitly expanded as

$$\text{select}(U) = \sum_{\gamma=1}^{\Gamma} |\gamma\rangle^{\otimes \Gamma} \langle \gamma|_C^{\otimes \Gamma} \otimes \left(\sum_{l \in v_{\gamma}} |l\rangle \langle l|_{\gamma} \otimes \frac{H_l}{\|H_l\|} \right) + \sum_{e=1}^{|\mathcal{E}|} |e\rangle^{\otimes \Gamma} \langle e|_C^{\otimes \Gamma} \otimes \frac{H_e}{\|H_e\|}. \quad (\text{B10})$$

It should be noted that we have projected the ancillary system onto its encoding basis $\{|0\rangle^{\otimes \Gamma}, |1\rangle^{\otimes \Gamma}, \dots, |J\rangle^{\otimes \Gamma}\}$. Now the d-BE of H gives

$$\left(\langle 0|_C G_C^{\dagger} \bigotimes_{\gamma=1}^{\Gamma} \langle 0|_{\gamma} G_{\gamma}^{\dagger} \right) \text{select}(U) \left(G_C |0\rangle_C \bigotimes_{\gamma=1}^{\Gamma} G_{\gamma} |0\rangle_{\gamma} \right) = \frac{H}{\alpha}. \quad (\text{B11})$$

Note that the implementations of d-BE of H have the same denominator $\alpha = \sum_{\gamma=1}^{\Gamma} \|H_{\gamma}\|_1 + \sum_{e \in \mathcal{E}} \|H_e\|$ as standard BE [36]. Expanding H may help us better understand this result. It can be seen that

$$H = \sum_{\gamma=1}^{\Gamma} H_{\gamma} + \sum_{e \in \mathcal{E}} H_e \equiv \sum_{\gamma=1}^{\Gamma} \|H_{\gamma}\|_1 \frac{H_{\gamma}}{\|H_{\gamma}\|_1} + \sum_{e \in \mathcal{E}} H_e \equiv \left(\sum_{\gamma=1}^{\Gamma} \|H_{\gamma}\|_1 + \sum_{e \in \mathcal{E}} \|H_e\| \right) \frac{H}{\sum_{\gamma=1}^{\Gamma} \|H_{\gamma}\|_1 + \sum_{e \in \mathcal{E}} \|H_e\|}. \quad (\text{B12})$$

The d-BE must be normalized with respect to the partition of H and the local BE for H_{γ} . Specifically, the auxiliary states of each local BE for H_{γ} are normalized, with the normalization coefficients equal to $\|H_{\gamma}\|_1$. The above equation shows that it is decomposed in a way that ensures the normalization of the coefficients for all local nodes, while also normalizing the coefficients of the control nodes (with the normalization coefficient being $\|H\|_1$).

Quantum communication complexity of d-BE. In the d-BE protocol described above, the quantum communication cost arises from the process of distributing the quantum state Eq. (B9), which requires the transmission of $\Gamma \log(|\mathcal{E}| + \Gamma)$ qubits. In addition, after executing $\text{select}(U)$, the control node needs to perform the $G_C^{\dagger}$ operation, thus requiring each node to resend the corresponding $\log(|\mathcal{E}| + \Gamma)$ qubits back to the control node. Thus the overall quantum communication cost of d-BE is $2\Gamma \log(|\mathcal{E}| + \Gamma)$, i.e., the quantum communication complexity is $\mathcal{O}(\Gamma \log(|\mathcal{E}| + \Gamma))$.

3. Distributed reflection operation (d-RO)

In the post-Trotter quantum simulation protocols, a crucial element besides BE is the reflection operation (RO) to amplify of the success probability of the algorithm. The reflection operation acting on the ancillary system is defined as $R = I - (1 - e^{-i\phi}) |0\rangle \langle 0|$. In the LCU and Grover's algorithm, one typically sets $\phi = \pi$, whereas in QSP the setting

of $\{\phi\}$ is dependent on the objective function. Our goal is to implement distributed reflection operation (denoted **d-R0**) in a quantum network (Γ nodes and a control node), that is

$$R = I - (1 - e^{-i\phi}) |0\rangle^{\otimes \Gamma} \langle 0|_C^{\otimes \Gamma} \otimes \bigotimes_{\gamma=1}^{\Gamma} |0\rangle\langle 0|_{\gamma} = I - (1 - e^{-i\phi}) |0\rangle\langle 0|_C \otimes \bigotimes_{\gamma=1}^{\Gamma} |0\rangle\langle 0|_{\gamma}, \quad (\text{B13})$$

where $|0\rangle_C = |i\rangle_C^{\otimes \Gamma}$. The reflection operation essentially performs a phase gate on the target component $|0\rangle$ (acting on $\Gamma \log |\mathcal{E}| + \sum_{\gamma=1}^{\Gamma} \log L_{\gamma}$ qubits of overall ancillary systems), which may require $\mathcal{O}(1)$ multi-qubit Toffoli gates to implement. For performing this reflection operation R , a naive approach is for each node to transfer its local ancillary state to the control node with the quantum communication complexity $\mathcal{O}(\sum_{\gamma=1}^{\Gamma} \log L_{\gamma})$. Obviously, the communication cost is large if each $L_{\gamma} \gg |\mathcal{E}| + \Gamma$. A smarter approach is to introduce auxiliary qubits and realize this multi-qubit reflection operation by cascading a series of Toffoli gates, such as the method in [78]. This approach requires the layout of the circuit to Γ nodes, and the communication complexity may be $\mathcal{O}(\Gamma)$. However, as mentioned above, the operation of cascading in fact increases the overall circuit depth. Considering the parallel nature of distributed computing, here we propose a **d-R0** based on nested d-LCUs in a network, which has the same communication complexity while having a shallower circuit depth compared to the former.

Here we give a concrete construction of **d-R0** in a quantum network with $\Gamma + 1$ nodes. First, we introduce the local reflection operation of γ -th node, which is defined as $R_{\gamma} = 2|0\rangle\langle 0|_{\gamma} - I_{\gamma}$. Similarly, the local reflection operation of the control node is $R_C = 2|0\rangle\langle 0|_C - I_C$. Moving the identity operator I_{γ} and I_C from the right to the left, we get

$$\frac{R_{\gamma} + I_{\gamma}}{2} = |0\rangle\langle 0|_{\gamma}, \quad \frac{R_C + I_C}{2} = |0\rangle\langle 0|_C. \quad (\text{B14})$$

Now we rewrite the R operation as

$$R = I - (1 - e^{-i\phi}) \left(\frac{R_C + I_C}{2} \right) \otimes \bigotimes_{\gamma=1}^{\Gamma} \left(\frac{R_{\gamma} + I_{\gamma}}{2} \right) = I - (1 - e^{-i\phi}) \bigotimes_{\gamma=0}^{\Gamma} \left(\frac{R_{\gamma} + I_{\gamma}}{2} \right). \quad (\text{B15})$$

For ease of expression, we have set the control node as node 0 i.e. $\gamma = 0$. Since R is unitary, and the above equation shows that R can be rewritten as a linear sum of unitary operations, one may introduce a new auxiliary system to implement **d-R0** via LCU. We can use the nested LCU technique to achieve this operation, where the auxiliary system of the control node is only 2-dimensional. Specifically, we introduce a new ancillary qubit at the control node and each local node, respectively, i.e. $|0\rangle_{C_b} \otimes \bigotimes_{\gamma=0}^{\Gamma} |0\rangle_{\gamma_b}$. These ancillary registers are prepared to

$$\left(G_{C_b} \otimes \bigotimes_{\gamma=1}^{\Gamma} H_{\gamma_b} \right) |0\rangle_{C_b} \otimes \bigotimes_{\gamma=0}^{\Gamma} |0\rangle_{\gamma_b} = \frac{1}{\sqrt{1 + 2 \sin \frac{\phi}{2}}} \left(|0\rangle + \sqrt{2 \sin \frac{\phi}{2}} |1\rangle \right)_{C_b} \otimes \bigotimes_{\gamma=0}^{\Gamma} |+\rangle_{\gamma_b}, \quad (\text{B16})$$

where all H_{γ_b} are Hadamard gate and $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. Note that $-(1 - e^{-i\phi}) = -ie^{-i\frac{\phi}{2}} 2 \sin \frac{\phi}{2}$ and we have assumed $\phi \in [0, \pi/2]$. Then, a single qubit phase gate $U_{\phi} = |0\rangle\langle 0| + |1\rangle\langle 1| e^{i(-\frac{\phi}{2} + \frac{3\pi}{2})}$ is implemented on the ancillary qubit C_b , and the ancillary register state (for $\{C_b, \gamma_b\}$) becomes

$$\frac{1}{\sqrt{1 + 2 \sin \frac{\phi}{2}}} \left(|0\rangle - ie^{-i\frac{\phi}{2}} \sqrt{2 \sin \frac{\phi}{2}} |1\rangle \right)_{C_b} \otimes \bigotimes_{\gamma=0}^{\Gamma} |+\rangle_{\gamma_b}. \quad (\text{B17})$$

Now we define the effective $\text{select}(U_R)$ for **d-R0** as

$$\text{select}(U_R) = |0\rangle\langle 0|_{C_b} \otimes I + |1\rangle\langle 1|_{C_b} \otimes \bigotimes_{\gamma=1}^{\Gamma} (|0\rangle\langle 0|_{\gamma_b} \otimes I_{\gamma} + |1\rangle\langle 1|_{\gamma_b} \otimes R_{\gamma}). \quad (\text{B18})$$

We have the **d-R0** as follows,

$$\left(G_{C_b} \otimes \bigotimes_{\gamma=1}^{\Gamma} H_{\gamma_b} \right)^{\dagger} \text{select}(U_R) U_{\phi} \left(G_{C_b} \otimes \bigotimes_{\gamma=0}^{\Gamma} H_{\gamma_b} \right) = \frac{|0\rangle_{C_b, \gamma_b} \langle 0|_{C_{\gamma}, \gamma_b} \otimes R}{1 + 2 \sin \frac{\phi}{2}} + \mathcal{R}, \quad (\text{B19})$$

where $|0\rangle_{C_b, \gamma_b} = |0\rangle_{C_b} \otimes \bigotimes_{\gamma=0}^{\Gamma} |0\rangle_{\gamma_b}$ and $\mathcal{R}$ is orthogonal to the subspace $\text{span}\{|0\rangle\langle 0|_{C_b, \gamma_b} \otimes \mathcal{H}_{tar}\}$. The success probability of R can be seen to depend on the target phase ϕ . According to oblivious amplitude amplification, one needs to query the whole unitary $\mathcal{O}\left(\sqrt{1 + 2\sin\frac{\phi}{2}}\right)$ times to guarantee that the success probability of R is 1. Remarkably, our scheme can be generalized to the distributed phase operation of any component of interest, i.e. $I - (1 - e^{-i\phi})|j\rangle\langle j|$.

Quantum communication complexity of d-RO. In general, the query complexity of implementing **d-RO** (R) is $\mathcal{O}(\sqrt{1 + 2\sin\frac{\phi}{2}}) = \mathcal{O}(1)$ since $1 \leq 1 + 2\sin\frac{\phi}{2} \leq 3$ ($\phi \in [0, \pi/2]$). In the above process of performing **d-RO**, only the control register C_b is needed to be distributed into the whole network (Γ nodes). That is, for an $\mathcal{O}(1)$ query of the oracle of **d-RO**, the quantum communication complexity is $\mathcal{O}(\Gamma)$. Without loss of generality, $|i\rangle_{C_b}$ can be defined as $|i\rangle_{C_b} = |i\rangle \otimes |i\rangle^{\otimes \Gamma}$, where each qubit in $|i\rangle^{\otimes \Gamma}$ should be individually sent to Γ nodes. In this setting, the quantum communication complexity for **d-RO** is still $\mathcal{O}(\Gamma)$ but with lower circuit depth.

4. Distributed truncated Taylor series (d-TS)

In this section, we would like to extend the truncated Taylor series (TS) [33] to its distributed version (denoted **d-TS**). Suppose $\left\|e^{-iHt/r} - \sum_{k=0}^K \frac{(-i)^k (t/r)^k}{k!} H^k\right\| \leq \epsilon/r$. Given $H = \sum_l H_l = \sum_{\gamma=1}^{\Gamma} H_{\gamma} + \sum_{e \in \mathcal{E}} H_e$, we denote the norm of a Hamiltonian term by $\alpha_l := \|H_l\|$ and $\alpha := \sum_l \alpha_l = \|H\|_1$ where $\|H\|_1 = \sum_j \|H_j\|$. For accuracy ϵ/r , the required order of Taylor expansion must be $K = \mathcal{O}\left(\frac{\log(\alpha t/\epsilon)}{\log \log(\alpha t/\epsilon)}\right)$ [33]. Expanding $e^{iHt/r} \approx \sum_{j=1}^J \beta_j \bigotimes_1^{\Gamma} U_j^{(\gamma)}$ based on truncated TS and one can implement quantum simulation of truncated TS using d-LCU with algorithm error ϵ/r .

Here we generalize the unary-encoding setting of TS method [33] to distributed case. The idea for unary-encoding for TS is intuitive, which essentially prepares an auxiliary system applying control operations allowing the linear combinations different H^k with coefficient $\frac{(it)^k}{k!}$. We present the quantum circuit for **d-TS** with unary encoding in Fig. 6. Specifically, the overall ancillary systems prepare

$$B_{\text{unary}} |0\rangle = G_{\text{TS}} \otimes \bigotimes_{i=1}^K G_{(i)} |0\rangle = \frac{1}{\sqrt{\alpha_{\text{TS}}}} \sum_{k=0}^K \sqrt{\frac{(\alpha t/r)^k}{k!}} |1^k 0^{K-k}\rangle \otimes \bigotimes_{i=1}^K \sum_{l=1}^L \frac{1}{\sqrt{\alpha}} \sqrt{\|H_l\|} |l\rangle_i, \quad (\text{B20})$$

where $\alpha_{\text{TS}} = \sum_{k=0}^K \frac{(\alpha t/r)^k}{k!}$ and $\alpha = \sum_l^L \|H_l\|$. Note that here $G_{(i)} = G_C \bigotimes_{\gamma=1}^{\Gamma} G_{\gamma}$ is the gate to encode all coefficients of H which is exactly equal to G in Appendix B 2. The $\text{select}(U_{\text{unary}})$ is given as

$$\text{select}(U_{\text{unary}}) = \bigotimes_{i=1}^K (|0\rangle\langle 0|_i \otimes I - i |1\rangle\langle 1|_i \otimes \text{select}(U_i)), \quad (\text{B21})$$

where $\text{select}(U_i) = \sum_l^L |j\rangle\langle j|_i \otimes \frac{H_l}{\|H_l\|_s}$ (which is equivalent to Eq. (B10)). Now we define $W_{\text{unary}} := B_{\text{unary}}^{\dagger} \otimes I_s(\text{select}(U_{\text{unary}})) B_{\text{unary}} \otimes I_s$ and then

$$W_{\text{unary}} |0\rangle |\psi\rangle_s = \frac{1}{\alpha_{\text{TS}}} |0\rangle \otimes \left(\sum_{k=0}^K \frac{(-i)^k (t/r)^k}{k!} H^k |\psi\rangle \right) + |\phi\rangle, \quad (\text{B22})$$

where $H = \sum_l^L H_l$. For oblivious amplitude amplification [56], one may set $\alpha_{\text{TS}} = e^{t/r\alpha} = 2$. That is $t/r = \ln 2/\alpha$. Specifically, one can approximate $|0\rangle \otimes e^{-iHt/r} (\bigotimes_{\gamma=1}^{\Gamma} |\psi_j\rangle)$ by

$$-W_{\text{unary}} R W_{\text{unary}}^{\dagger} R W_{\text{unary}} |0\rangle \otimes \left(\bigotimes_{\gamma=1}^{\Gamma} |\psi_j\rangle \right) = |0\rangle \otimes \sum_{k=0}^K \frac{(-i)^k (t/r)^k}{k!} H^k \left(\bigotimes_{\gamma=1}^{\Gamma} |\psi_j\rangle \right) \quad (\text{B23})$$

where $|\psi(t=0)\rangle_s = \bigotimes_{\gamma=1}^{\Gamma} |\psi_j\rangle$ is the product state. By repeating the above process, the (distributed) quantum dynamics $U \approx (e^{-iHt/r})^r = e^{-iHt}$ can be implemented.

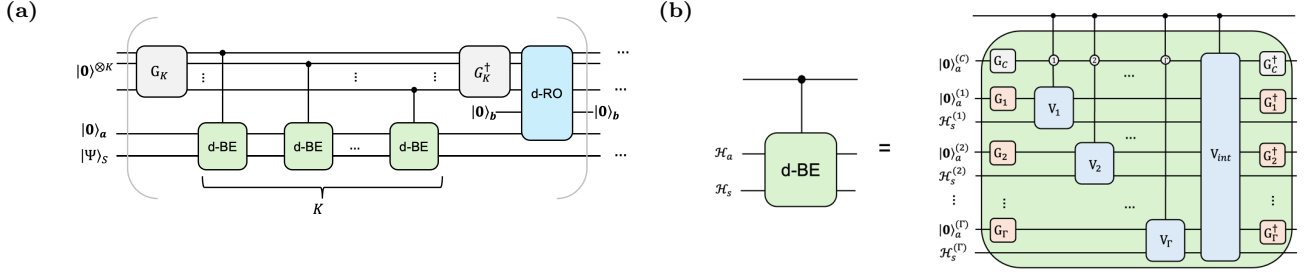


FIG. 6. (a) The oracle operator for **d-TS** circuit. G_{TS} is used to prepare $\frac{1}{\sqrt{\alpha_{TS}}} \sum_{k=0}^K \sqrt{\frac{(\alpha t/r)^k}{k!}} |1^k 0^{K-k}\rangle$, which is needed to be transmitted to the entire network. (b) The quantum circuit for controlled **d-BE**. Here $\mathcal{H}_{anc}$ and $\mathcal{H}_{sys}$ represent all ancillary systems $\mathcal{H}_{anc}^{(C)} \otimes \mathcal{H}_{anc}^{(1)} \otimes \mathcal{H}_{anc}^{(2)} \dots \otimes \mathcal{H}_{anc}^{(\Gamma)}$ and the systems of interest $\mathcal{H}_{sys}^{(1)} \otimes \mathcal{H}_{sys}^{(2)} \dots \otimes \mathcal{H}_{sys}^{(\Gamma)}$.

Theorem B.1 (Quantum communication complexity of **d-TS**, Restatement Theorem 2 in maintext). *Given a Γn -qubit Hamiltonian $H = \sum_{\gamma=1}^{\Gamma} H_{\gamma} + \sum_{e \in \mathcal{E}} H_e$ and evolution time t , the quantum communication complexity of the distributed TS protocol (**d-TS**) for quantum simulation with accuracy ϵ is*

$$\mathcal{O}\left(\alpha \Gamma \log(|\mathcal{E}| + \Gamma) \frac{\log(\alpha t/\epsilon)}{\log \log(\alpha t/\epsilon)}\right), \quad (\text{B24})$$

where Γ is the number of nodes, $|\mathcal{E}|$ is the number of the interaction terms, and $\alpha = \|H\|_1$.

Proof of Theorem B.1. Given an Γn -qubit Hamiltonian $H = \sum_{\gamma=1}^{\Gamma} H_{\gamma} + \sum_{e \in \mathcal{E}} H_e$ and evolution time t , the quantum communication complexity of **d-TS** for quantum simulation is the query complexity multiply the quantum communication complexity of each oracle (query). As we shown in Fig. 6, a single query consists K controlled **d-BE** of Hamiltonian H and a single **d-RO** for π phase. Based on our analysis before, the quantum communication complexity for **d-BE** and **d-RO** are $\mathcal{O}(\Gamma \log(|\mathcal{E}| + \Gamma))$ and $\mathcal{O}(\Gamma)$ respectively. Compared to **d-BE**, here we need to consider the quantum communication complexity of controlled **d-BE**. The quantum circuit of controlled **d-BE** is presented in Fig. 6 (b). One can see that the introduced control qubit needs to be sent to each node, which in turn performs the corresponding control operation, and hence it introduces an additional quantum communication cost of Γ qubits. Thus the quantum communication complexity for controlled **d-BE** is also $\mathcal{O}(\Gamma \log(|\mathcal{E}| + \Gamma))$. Overall, the quantum communication complexity of **d-TS** is $\mathcal{O}(\alpha t (1 + \log(|\mathcal{E}| + \Gamma)) \Gamma K) = \mathcal{O}(\alpha t \log(|\mathcal{E}| + \Gamma) \Gamma K)$ where Γ is the number of nodes, $|\mathcal{E}|$ is the number of the interaction terms, $\alpha := \|H\|_1$ and $K = \mathcal{O}\left(\frac{\log(\alpha t/\epsilon)}{\log \log(\alpha t/\epsilon)}\right)$. $\square$

5. Distributed quantum signal processing (**d-QSP**)

Quantum signal processing (QSP), sometimes it may be called qubitization is a powerful technique that allows one to perform a function of high dimensional matrices [35, 36]. The key idea of qubitization is to divide the high-dimensional block encoding unitary into two-dimensional subspace, which can be considered an effective qubit.

QSP can implement a wide range of polynomial functions, what we are interested in here is how it can be applied to quantum (Hamiltonian) simulations. Specifically, QSP utilizes the Jacobi-Anger expansion to implement $e^{-iHt} = \sum_{k=-\infty}^{\infty} i^k J_k(t) e^{ik \cos^{-1} H}$, by querying multiple H and phase gates, where $J_k(t)$ represents the Bessel function of the first kind. By QSP, the query complexity needed to approximate e^{-iHt} within error ϵ is $\mathcal{O}\left(\alpha t + \frac{\log(1/\epsilon)}{\log \log(1/\epsilon)}\right)$. Compared to the Taylor truncated method, QSP may have a lower query complexity. One may find more details in Ref [36]. Here we generalize QSP to a quantum network (denoted **d-QSP**).

The QSP algorithm consists of the block-encoding of Hamiltonian and the reflection and phase operations acting on the auxiliary system [36]. Similarly, as shown in Fig. 7, **d-QSP** shall consist of **d-BE** and corresponding reflection acting on the auxiliary system acting on the auxiliary system. This construction of **d-QSP** is simple and it follows the same query setup as the non-distributed one. The difference between them mainly stems from the partition of Hamiltonian in the network, i.e., the interaction graphs which in turn make the design of **d-BE** of H and **d-RO** different from the original one. By the following Lemma of QSP proposed by Low and Chuang [36], we can obtain the quantum communication complexity of **d-QSP**.

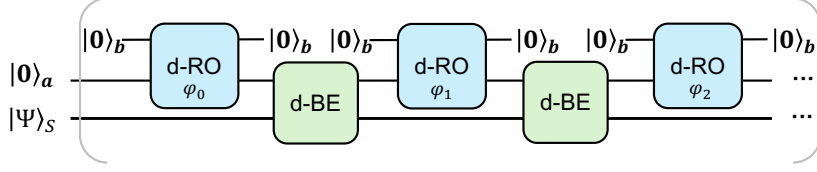


FIG. 7. The quantum circuit for **d-QSP**. Here **d-BE** denotes the oracle of block encoding of the target matrix, i.e. the Hamiltonian H . The **d-RO** may act as a specific phase on the component $|0\rangle_a$ of (all) ancillary system a .

Lemma B.1 (Hamiltonian Simulation by Qubitization [36]). *Given access to the oracles of $\langle 0|_a(G^\dagger \otimes I)$ $\text{select}(U)$ ($G \otimes I$) $|0\rangle_a = \frac{H}{\alpha}$ specifying a Hamiltonian $H = \sum_i^d \alpha_i U_i$ where $\alpha = \sum_i |\alpha_i|$, time evolution by H can be simulated for time t and error ϵ with $\mathcal{O}(\alpha t + \log(1/\epsilon))$ queries.*

Theorem B.2 (Quantum communication complexity of **d-QSP**, Theorem 3 in maintext). *Given a Γn -qubit Hamiltonian $H = \sum_{\gamma=1}^\Gamma H_\gamma + \sum_{e \in \mathcal{E}} H_e$ and evolution time t , the quantum communication complexity of the distributed QSP protocol (**d-QSP**) for quantum simulation with accuracy ϵ is*

$$\mathcal{O}(\Gamma \log(|\mathcal{E}| + \Gamma)(\alpha t + \log(1/\epsilon))), \quad (\text{B25})$$

where Γ is the number of nodes, $|\mathcal{E}|$ is the number of the interaction terms, and $\alpha = \|H\|_1$.

Proof of Theorem B.2. As we analyzed before, for $H = \sum_{\gamma=1}^\Gamma H_\gamma + \sum_{e \in \mathcal{E}} H_e$, the quantum communication complexity of the oracles of **d-BE** and **d-RO** are $\mathcal{O}(\Gamma \log(|\mathcal{E}| + \Gamma))$ (see Appendix B 2) and $\mathcal{O}(\Gamma)$ (see Appendix B 3) respectively. Therefore, for a single query of **d-QSP**, the quantum communication complexity is still $\mathcal{O}(\Gamma \log(|\mathcal{E}| + \Gamma))$. The quantum communication complexity of **d-QSP** should be the query complexity multiplied by the quantum communication complexity of each **d-BE** and **d-RO**, i.e., $\mathcal{O}(\Gamma \log(|\mathcal{E}| + \Gamma)(\alpha t + \log(1/\epsilon)))$. $\square$

Similar to **d-TS**, the quantum communication complexity of **d-QSP** is linear with t and proportional to the number of nodes Γ and the number of interaction terms across nodes $|\mathcal{E}|$.

Appendix C: Applications in distributed quantum simulation

In this section, we analyze the quantum communication complexity of different distributed quantum simulation protocols for some common physical systems. Beyond quantum simulation, we show that the distributed (quantum network) protocols can be applied to other important quantum algorithms, such as quantum phase estimation and Grover search, with low quantum communication.

1. Lattice nearest-neighbor Hamiltonians

We begin with the one-dimensional (1D) lattice spin Hamiltonian with the nearest-neighbor (NN) interaction $H_{\text{NN}} = \sum_{j=1}^{n-1} H_{j,j+1}$ where $H_{j,j+1}$ is a Hermitian operator that acts nontrivially only on qubits j and $j+1$.

Lemma C.1 (Product formula for NN Hamiltonians [52, 79]). *Let H be a Hamiltonian with n qubits. Let $\tilde{U}_p$ be a p th-order product formula. The number of Trotter steps for simulation accuracy ϵ is $r = \mathcal{O}(n^{1/p} t^{1+1/p} / \epsilon^{1/p})$.*

The most straightforward method for quantum simulation of an NN Hamiltonian in a quantum network is to implement the PF distributively as in Appendix A. Firstly, we regroup the (Γn -qubit) NN Hamiltonian according to its interaction (the terms across nodes) and network topology of Γ nodes, i.e., $H = H_0 + H_{\mathcal{E}}$ where $H_0 = \sum_{\gamma=1}^\Gamma H_\gamma$ and $H_{\mathcal{E}} = \sum_{e \in \mathcal{E}} H_e$. The terms $\text{supp}(H_\gamma) \subseteq N_\gamma$ can be implemented (e.g. by the high-order PFs) on the γ th node locally to arbitrary precision without communication. The interaction terms H_e represent the terms whose support is across two neighboring nodes. Note that given a Γ -partite quantum network to simulate a 1D NN Hamiltonian with Γn qubits, the number of interaction terms is $|\mathcal{E}| = \mathcal{O}(\Gamma)$. Therefore, we have the following proposition.

Models \ methods	p th-order d-PF	d-TS	d-QSP
General H	$\mathcal{O}(\Gamma \tilde{\alpha}_{\text{comm},p}^{1/p} \mathcal{E} t^{1+1/p} / \epsilon^{1/p})$	$\mathcal{O}\left(\log(\mathcal{E} + \Gamma) \Gamma \alpha t \frac{\log(\alpha t / \epsilon)}{\log \log(\alpha t / \epsilon)}\right)$	$\mathcal{O}(\log(\mathcal{E} + \Gamma) \Gamma (\alpha t + \log(1/\epsilon)))$
k -local	$\mathcal{O}\left(\min(k, \Gamma) (\Gamma n)^k \ H\ _1 \alpha^{1/p} t^{1+1/p} / \epsilon^{1/p}\right)$	$\mathcal{O}\left(k \log(\Gamma n) \Gamma \alpha t \frac{\log(\alpha t / \epsilon)}{\log \log(\alpha t / \epsilon)}\right)$	$\mathcal{O}(k \log(\Gamma n) \Gamma (\alpha t + \log(1/\epsilon)))$
1D NN Lattice	$\mathcal{O}(\Gamma^{1+1/p} t^{1+1/p} / \epsilon^{1/p})$	$\mathcal{O}(\log(\Gamma) \Gamma \alpha t \frac{\log(\alpha t / \epsilon)}{\log \log(\alpha t / \epsilon)})$	$\mathcal{O}(\log(\Gamma) \Gamma (\alpha t + \log(1/\epsilon)))$

TABLE I. Quantum communication complexity of three quantum simulation protocols (i.e., distributed product formula (d-PF), truncated Taylor series method (d-TS), quantum signal processing (d-QSP)) for different types of Γn -qubit physical systems. $\tilde{\alpha}_{\text{comm},p}$ is the norm of the nested commutator of terms in H defined in Theorem 1. Γ is the number of nodes, $\alpha = \|H\|_1$ and $|\mathcal{E}|$ is the number of the induced interaction terms that act on multiple nodes. $\|H\|_1 = \max_l \max_{j_l} \sum_{j_1, \dots, j_{l-1}, j_{l+1}, \dots, j_k} \|H_{j_1, \dots, j_k}\|$ is the induced 1-norm [52]. For the k -local case, we replace Γ with $\min(k, \Gamma)$ inside d-PF, since the support of the interaction terms under this model is at most k . The 1D lattice nearest-neighbor (NN) Hamiltonian is a special case of k -local model, i.e. $k = 2$. We color the same scaling to make it easier for readers.

Proposition C.1 (d-PF for NN Hamiltonians). *Given a Γ -partite quantum network, let H be a one-dimensional lattice Hamiltonian with Γn qubits, and $\tilde{U}_p$ be a p th-order product formula, the number of Trotter steps (round of quantum communications) for simulation accuracy ϵ is $r = \mathcal{O}((\Gamma)^{1/p} t^{1+1/p} / \epsilon^{1/p})$. Then, the (total) quantum communication complexity of the d-PF protocol for NN Hamiltonians is $\mathcal{O}(\Gamma (\Gamma)^{1/p} t^{1+1/p} / \epsilon^{1/p})$.*

The proof of this proposition is intuitive. Childs and Su [79] made use of even-odd ordering to rewrite the Hamiltonian as $H = H_{\text{even}} + H_{\text{odd}}$ in which each term inside H_{even} or H_{odd} is pairwise commutative. For 1D NN Hamiltonian implemented by a star network, the clustered Hamiltonian can also be written as $H_0 = H_{\text{even}}$ and $H_{\mathcal{E}} = H_{\text{odd}}$. Since $[H_\gamma, H_{\gamma'}] = 0$ for $\gamma \neq \gamma'$ and one may package the interaction terms between the γ -th node and the $\gamma + 1$ -th node as $H_{\gamma, \gamma+1}$ so that $H_{\mathcal{E}} = \sum_{\gamma=1}^{\Gamma-1} H_{\gamma, \gamma+1}$. Thus it suffices to take $r = \mathcal{O}(\Gamma^{1/p} t^{1+1/p} / \epsilon^{1/p})$ to bound the Trotter error (we have replaced n in Lemma C.1 with Γ). Similarly, for general case where $|\mathcal{E}| = \mathcal{O}(\Gamma)$, the complexity of post-Trotter methods is as follows.

Proposition C.2 (d-TS for NN Hamiltonians). *Given a Γ -partite quantum network, let H be the one-dimensional lattices Hamiltonian with Γn qubits, the quantum communication complexity of the d-TS protocol for NN Hamiltonians is $\mathcal{O}\left(\Gamma \alpha t \log(\Gamma) \frac{\log \frac{\alpha t}{\epsilon}}{\log \log \frac{\alpha t}{\epsilon}}\right)$ where ϵ is the simulation accuracy and $\alpha := \|H\|_1$.*

Proposition C.3 (d-QSP for NN Hamiltonians). *Given a Γ -partite quantum network, let H be the one-dimensional lattices Hamiltonian with Γn qubits, the quantum communication complexity of the d-QSP protocol for NN Hamiltonians is $\mathcal{O}(\Gamma \log(\Gamma) (\alpha t + \log(1/\epsilon)))$ where ϵ is the simulation accuracy and $\alpha := \|H\|_1$.*

2. k -local Hamiltonians

The k -local Hamiltonian is a common model in the context of quantum computation and simulation. Each term of a k -local Hamiltonian $H_k = \sum_{j_1, \dots, j_k} H_{j_1, \dots, j_k}$ acts non-trivially on at most k qubits where each term $H_{j_1, \dots, j_k}$ has support $\{j_1, \dots, j_k\}$. We first introduce a lemma of the product formula for k -local Hamiltonians.

Lemma C.2 (PF for k -local Hamiltonians [52]). *Let H be a k -local Hamiltonian with n qubits. Then, the nested commutator of a p th-order product formula is*

$$\tilde{\alpha}_{\text{comm}} = \sum_{\gamma_1, \gamma_2, \dots, \gamma_{p+1}} \|[H_{\gamma_{p+1}}, \dots, [H_{\gamma_2}, H_{\gamma_1}]]\| = \mathcal{O}(\|H\|_1^p \|H\|_1), \quad (\text{C1})$$

where $\|H\|_1^p = \max_l \max_{j_l} \sum_{j_1, \dots, j_{l-1}, j_{l+1}, \dots, j_k} \|H_{j_1, \dots, j_k}\|$ is the induced 1-norm. The number of Trotter steps for achieving simulation accuracy ϵ is $r = \mathcal{O}(\|H\|_1^p t^{1+1/p} / \epsilon^{1/p})$.

With the Trotter steps for k -local Hamiltonians, we have the following quantum communication complexity of d-PF.

Proposition C.4 (d-PF for k -local Hamiltonians). *Given a Γ -partite quantum network, and let $H = H_0 + \sum_{e \in \mathcal{E}} H_e = \sum_{e=0}^{|\mathcal{E}|} H_e$, (where $H_0 = \sum_{\gamma=1}^{\Gamma} H_\gamma$ and $|\mathcal{E}|$ is the number of the induced edge terms) be a k -local Hamiltonian with Γn qubits. Let $\tilde{U}_p$ be a p th-order product formula. Then, the nested commutator*

$$\tilde{\alpha}_{\text{comm}} = \sum_{e_1, e_2, \dots, e_{p+1}=0}^{|\mathcal{E}|} \|[H_{e_{p+1}}, \dots, [H_{e_2}, H_{e_1}]]\| = \mathcal{O}(\|H\|_1^p \|H\|_1), \quad (\text{C2})$$

where $\|H\|_1^p = \max_l \max_{j_l} \sum_{j_1, \dots, j_{l-1}, j_{l+1}, \dots, j_k} \|H_{j_1, \dots, j_k}\|$ is the induced 1-norm. The number of Trotter steps (round of quantum communications) for simulation accuracy ϵ is $r = \mathcal{O}(\|H\|_1^p \|H\|_1^{1/p} t^{1+1/p} / \epsilon^{1/p})$. Then, the (total) quantum communication complexity of the d-PF protocol for k -local Hamiltonian is

$$\mathcal{O}(\min(k, \Gamma)(\Gamma n)^k \|H\|_1^p \|H\|_1^{1/p} t^{1+1/p} \epsilon^{1/p}). \quad (\text{C3})$$

Proof. In each segment, $|\mathcal{E}|$ gates for interaction crossing k nodes must be implemented with quantum communication. Suppose each node has n qubits, then the number of interaction terms for global Γn -qubit k -local Hamiltonian is $|\mathcal{E}|_k = \mathcal{O}((\Gamma n)^k)$. In general, for non-distributed cases, the number of terms for Γn -qubit with k -local Hamiltonian is also $\mathcal{O}((\Gamma n)^k)$ [52]. Considering the distributed case, subtracting the term of each node local we still have $\mathcal{O}((\Gamma n)^k) - \mathcal{O}(\Gamma n^k) = \mathcal{O}((\Gamma n)^k)$. More specifically, for 2-local case ($\Gamma \geq k = 2$), $|\mathcal{E}|_k = c \binom{\Gamma}{2} \binom{n}{1}^2 = c \frac{\Gamma(\Gamma-1)}{2} n^2$, where c is a constant. For $\Gamma \geq k > 2$, the $|\mathcal{E}|$ is $\mathcal{O}((\Gamma n)^k)$. Since for d-PF, the segment number is $r = \mathcal{O}(\|H\|_1^p \|H\|_1^{1/p} t^{1+1/p} / \epsilon^{1/p})$, and for each interaction term (even if k is greater than Γ), it takes at most $\mathcal{O}(\min(k, \Gamma))$ communication to implement the corresponding exponential $e^{-i\delta t \|H_{e_m}\| \otimes_{j=1}^{\Gamma} (\otimes_{i \in v_j} \sigma_{z_i})}$. Therefore, the total quantum communication complexity of the d-PF protocol for k -local Hamiltonian is $\mathcal{O}(\min(k, \Gamma)(\Gamma n)^k \|H\|_1^p \|H\|_1^{1/p} t^{1+1/p})$. $\square$

For general case of k -local Hamiltonian which has $|\mathcal{E}|_k = \mathcal{O}((\Gamma n)^k)$, the quantum communication complexity of post-Trotter methods (d-TS and d-QSP) is directly given below.

Proposition C.5 (d-TS for k -local Hamiltonians). *Consider a k -local Hamiltonian H with Γn qubits in a Γ -partite quantum network, time t , and simulation accuracy ϵ . The total quantum communication complexity of d-TS is $\mathcal{O}\left(\alpha k \Gamma \log(\Gamma n) \frac{\log(\alpha t / \epsilon)}{\log \log(\alpha t / \epsilon)}\right)$ where $\alpha = \|H\|_1$.*

Proposition C.6 (d-QSP for k -local Hamiltonians). *Consider a k -local Hamiltonian H with Γn qubits in a Γ -partite quantum network, evolution time t , and simulation accuracy ϵ . The total quantum communication complexity of d-QSP is $\mathcal{O}(k \Gamma \ln(\Gamma n)(\alpha t + \log(1/\epsilon)))$ where $\alpha = \|H\|_1$.*

Appendix D: Extension to other distributed quantum algorithms

1. Distributed quantum phase estimation (d-QPE)

Quantum phase estimation (QPE) is a core procedure of many algorithms (e.g., Shor's algorithm [80]), which makes it significant in distributed quantum computing [81]. Our distributed protocols can be simply extended for distributed quantum phase estimation. Recall that quantum phase estimation has two registers, the first register is the control system and the phase information may be read out by measuring on a Fourier basis. While the second register is the system being used to execute the target unitary U .

In our distributed QPE (denoted d-QPE), different powers of U need to be implemented on a composite system of Γ nodes. In the quantum phase estimation algorithm, K is a constant number and assuming $U = \sum_{j=1}^J \beta_j \bigotimes_{\gamma=1}^{\Gamma} U_j^{(\gamma)}$, U^k can also be implemented directly with d-LCU (see Appendix B1). As shown in Fig. 8 (a), one may need to introduce additional K qubits at the control node to implement QPE. Recall that for d-LCU, one may perform $U = \sum_{j=1}^J \beta_j \bigotimes_{\gamma=1}^{\Gamma} U_j^{(\gamma)}$ with quantum communication complexity $\mathcal{O}(\sqrt{\sum_j |\beta_j|} \Gamma \log J)$. Clearly, for U^{2^K} , the communication complexity is $\mathcal{O}(2^K \sqrt{\sum_j |\beta_j|} \Gamma \log J)$. However, for QPE, controlled U is executed in the network, so an analysis of the communication complexity it may introduce is required.

Similar to controlled d-BE, here the communication complexity of a single controlled d-LCU of U introduced an additional (little) quantum communication overhead of Γ qubits since the control qubit needs to be sent to each node, which in turn performs the corresponding control operations. Thus, the quantum communication complexity of controlled d-LCU of U is still $\mathcal{O}(\sqrt{\sum_j |\beta_j|} \Gamma \log J)$. That is for a single controlled U^{2^K} , the quantum communication complexity is $\mathcal{O}(2^K \sqrt{\sum_j |\beta_j|} \Gamma \log J)$. Overall, one has the following proposition for general unitaries.

Proposition D.1 (d-QPE for general U). *Given a Γ -partite quantum network and $U = \sum_{j=1}^J \beta_j \bigotimes_{\gamma=1}^{\Gamma} U_j^{(\gamma)}$, the total quantum communication complexity of d-QPE is $\mathcal{O}\left(2^{2K} \sqrt{\sum_j |\beta_j|} \Gamma \log J\right)$, where Γ is the number of nodes and K is the number of decimals in the phase estimate.*

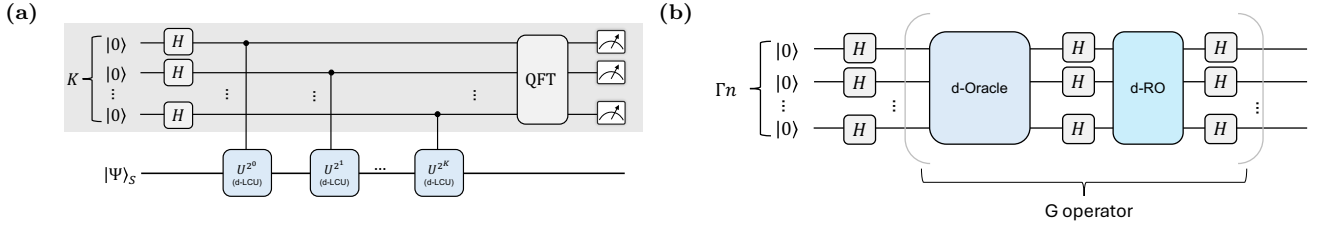


FIG. 8. (a) Distributed quantum phase estimation. There are K ancillary qubits at the control node and $|\psi_S\rangle$ is one of the eigenstates of U , which is pre-distributed to a Γ -partite quantum network. (b) Distributed Grover's algorithm based on the original Grover's algorithm. There are Γn qubits of data distributed in Γ nodes. Here the Oracle and reflection operations in Grover's algorithm are substituted with their distributed counterparts, d-Oracle and d-RO, respectively.

Similarly, we can consider the case where $U = e^{-iHt}$ is implemented by d-QSP for a Hamiltonian H of Γn qubits.

Proposition D.2 (d-QPE for dynamics of H). *Given a Γ -partite quantum network, a Hamiltonian in the form $H = \sum_{\gamma=1}^{\Gamma} H_{\gamma} + \sum_{e \in \mathcal{E}} H_e$ of Γn qubits, and a unitary $\tilde{U}$ satisfying $\|\tilde{U} - e^{-iHt}\| \leq \epsilon$, the total quantum communication complexity of d-QPE for $\tilde{U}$ is $\mathcal{O}(2^{2K} \Gamma \log(|\mathcal{E}| + \Gamma)(\alpha t + \log(1/\epsilon)))$, where Γ is the number of nodes, $|\mathcal{E}|$ is the number of interaction terms, $\alpha = \|H\|_1$ and K is the number of decimals in the phase estimate. The total error of d-QPE for the ideal $U = e^{-iHt}$ is $2^{2K} \epsilon$.*

2. Distributed Grover's algorithm (d-Grover)

Grover's algorithm is a quantum algorithm for searching an unsorted database with a speed-up over classical algorithms. While classical computation requires linear time to search for a specific item in a database, Grover's algorithm leverages the power of quantum computation to perform the task in square root time complexity, i.e., $\mathcal{O}(\sqrt{N})$, where N represents the number of items in the database. We note that there are different versions of distributed Grover's algorithm. Here our protocol (denoted **d-Grover**) refers specifically to the direct extension of the original Grover's algorithm where the Oracle and reflection operations are substituted with their distributed counterparts. We present the quantum circuit for **d-Grover** in Fig. 8 (b).

It can be seen that the core of Grover's algorithm is the execution of Oracle and reflection operations. The latter reflection operation we have already mentioned above can be generalized to d-RO. While Oracle can be considered as generalized reflection operations acting on a specific target component, i.e. $I - (1 - e^{-i\phi})|j\rangle\langle j|$. Thus in a quantum network with Γ nodes, it is straightforward to construct distributed Oracle (d-Oracle) based on d-RO. Recall that, in Appendix B 3, we have shown the quantum communication complexity for a general d-RO ($I - (1 - e^{-i\phi})|j\rangle\langle j|$) is $\mathcal{O}(\Gamma)$. Thus, for a single query of G operator (contains a d-Oracle and a d-RO), its quantum communication complexity is also $\mathcal{O}(\Gamma)$. Therefore, the total quantum communication complexity of distributed Grover's search should be $\mathcal{O}(\Gamma\sqrt{N})$.

Proposition D.3 (d-Grover in a Γ -partite network). *Given a Γ -partite quantum network, the total quantum communication complexity for d-Grover is $\mathcal{O}(\Gamma\sqrt{N})$, where Γ is the number of nodes and $N = 2^{\Gamma n}$ represents the number of items in the database.*

Appendix E: Generality of distributed protocols in quantum network with any topology

Here we show that for distributed protocols that require the assistance of control nodes, such as d-LCU, d-TS, d-QSP, d-QPE, and d-Grover and other such algorithms, their communication complexity is independent of the topology of the realistic quantum network. This conclusion is predicated on the fact that each node possesses $\log(n)$ auxiliary qubits in addition to n working qubits, and all auxiliary qubits of the whole network can be virtually and equivalently constructed as the centralized control node mentioned in the main text, through the quantum communication.

As shown in Fig. 9, there are two Γ -node quantum networks ($\Gamma = 6$) presented in (a) and (b) with different topologies. Each node has a primary quantum processor (n qubits) and a small ancillary system ($\mathcal{O}(\log(n))$ qubits), and any local operation can be implemented on the composite system of these two systems. The light blue box represents the primary quantum processor, while the small gray box represents the auxiliary quantum system.

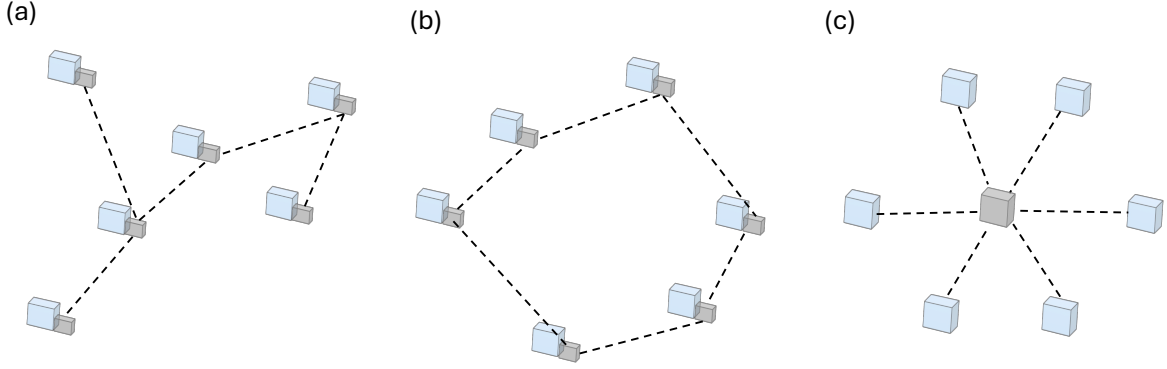


FIG. 9. Quantum network with various topologies. Each node has a primary quantum processor and a small ancillary system, and any local operation can be implemented on the composite system of these two systems. The light blue box represents the primary quantum processor, while the small gray box represents the auxiliary quantum system. The dashed line indicates a quantum connection, i.e. quantum channel. (a) and (b) can be virtually and equivalently constructed as (c), the Star network, through the quantum connection between ancillary systems.

As the auxiliary systems (small gray boxes) are connected via quantum channels, entanglement can be established between them via quantum communication. For example, if one of the nodes named node 1 has an auxiliary system preparing $G|\mathbf{0}\rangle = \sum_j \beta_j |j\rangle^{(1)} |j\rangle^{(1)}$, then it may transfer half of the qubits of state to the neighboring node 2 via quantum teleportation, at this point, node 1 and the neighboring node 2 shares the entangled state $\sum_j \beta_j |j\rangle^{(1)} |j\rangle^{(2)}$. For the information of the state of interest to be shared across the network, node 2 performs multiple local CNOT gates on the qubits on hand and other auxiliary qubits, thus preparing $\sum_j \beta_j |j\rangle^{(1)} |j\rangle^{(2)} |j\rangle^{(2)}$. Similarly, node 2 teleports half of these quantum states to node 3. And so on, eventually the entire network will share auxiliary states such as $\sum_j \beta_j |j\rangle^{(1)} |j\rangle^{(2)} \dots |j\rangle^{(\Gamma)}$, consistent with those prepared by central node in Fig. 9 (c) which is exactly we introduced in **d-BE**, **d-R0** and **d-LCU**.

Consequently, the above process can be considered equivalent to transforming a network of arbitrary topology into a virtual star network. Moreover, the quantum communication cost arises only from the two neighboring nodes, so the quantum communication complexity of the network constructing such a globally entangled auxiliary state is equal to the number of quantum connections (dash lines), which is proportional to the number of nodes. Similarly, its reverse process can also be implemented step by step. That is, the quantum communication complexity of an auxiliary system state sharing a full network is $\mathcal{O}(\Gamma)$, identical to the communication complexity required for the distribution of entanglement by the control nodes in a star network.

On the contrary, **d-PF** may require numbers of quantum teleportation or quantum state transfer to connect a certain two or more nodes each time it performs the evolution of certain interaction terms, such that its quantum communication complexity is somewhat dependent on the topology of the network.

Appendix F: Communication lower bound of distributed quantum simulation

The area of communication complexity deals with the scenario where multiple parties do computation collectively. Under this model, local computation is free, but communication is expensive and has to be minimized. Assume there are two separated parties, Alice and Bob who have a bit-string $\mathbf{y} \in \{0,1\}^n$ and $\mathbf{z} \in \{0,1\}^n$ respectively, they want to collectively evaluate a Boolean function $F(\mathbf{y}, \mathbf{z}) : \{0,1\}^n \times \{0,1\}^n \rightarrow \{0,1\}$ by local computation and communication with each other. Under the (standard) two-party quantum communication model, Alice and Bob each have a quantum computer and are allowed to communicate qubits (or to make use of the quantum correlations given by shared EPR pairs). This model was introduced by Yao [82] and has been studied extensively [41, 43]. The communication complexity of $F(\mathbf{y}, \mathbf{z})$ is the minimal amount of communication required for the worst-case input.

Definition 1 (Qcc_2). Under the standard two-party quantum communication model, the (*two-sided*) *bounded error quantum communication complexity* of a Boolean function $F(\mathbf{y}, \mathbf{z})$, denoted $\text{Qcc}_2(F)$, is the minimal number of qubits that Alice and Bob require to exchange to return $F(\mathbf{y}, \mathbf{z})$ with correct probability at least $2/3$ on the worst-case input.

To prove the lower bound of quantum communication complexity of distributed quantum simulation (d-QS), we utilize the distributed version of $\text{PARITY}_n(\mathbf{x})$ known as the INNERPRODUCT problem with known lower bound.

Problem 1 ($\text{IP}_n(\mathbf{y}, \mathbf{z})$). Under the standard two-party quantum communication model, the INNERPRODUCT (denoted IP_n) problem refers to evaluating the Boolean function $\text{IP}_n(\mathbf{y}, \mathbf{z}) := \bigoplus_{j=1}^n (y_j \wedge z_j) \equiv \mathbf{y} \cdot \mathbf{z} \bmod 2$ by communicating as few as possible (qu)bits between Alice and Bob who have the bit-string $\mathbf{y} \in \{0, 1\}^n$ and $\mathbf{z} \in \{0, 1\}^n$ respectively.

Lemma F.1 (Tight bound of IP_n). $\text{Qcc}_2(\text{IP}_n) = \Theta(n)$.

The intuition behind Lemma F.1 is by the following: one can use the protocol for INNERPRODUCT to transfer Alice's n -bit input to Bob, and then by Holevo's theorem which states that n qubits must be communicated to achieve this. We refer readers to [41, 60] for careful proof of this theorem. For Γ -partite lower bound, Le Gall and Suruga [61] proved the lower bound of $\text{Qcc}_2(\text{IP}_n)$ can be generalized from bipartite to Γ -partite case as $\Omega(\Gamma n)$.

Problem 2 (Γ -partite INNERPRODUCT $\text{IP}_{n,\Gamma}$). The Γ -partite INNERPRODUCT Problem $\text{IP}_{n,\Gamma} : \{0, 1\}^{\Gamma n} \rightarrow \{0, 1\}$ is defined as

$$\text{IP}_{n,\Gamma}(\mathbf{x}_1, \dots, \mathbf{x}_\Gamma) = \bigoplus_{j=1}^n (x_{1j} \wedge x_{2j} \cdots \wedge x_{\Gamma j}),$$

where $\mathbf{x}_i \in \{0, 1\}^n$ for $i = \{1, \dots, \Gamma\}$.

Lemma F.2 (Quantum communication complexity of $\text{IP}_{n,\Gamma}$ function [61]). *To compute $\text{IP}_{n,\Gamma}$ function with bounded-error using an oblivious quantum protocol, the number of qubits required to communicate in Γ partite is at least $\Theta(\Gamma n)$.*

Their lemma requires the communication model to be oblivious, which means the amount of communication exchanged between each pair of players at each round is fixed independently of the input before the execution of the protocol. Our setting satisfies this requirement.

To reduce IP_n to d-QS, we introduce a useful constructive tool called *circuit-to-Hamiltonian map* (or known as *perfect state transfer*) which maps a quantum circuit to a Hamiltonian evolution:

Lemma F.3 (Circuit-to-Hamiltonian map [21, 83, 84]). *Given a quantum circuit $U = U_N \dots U_1$ composed of N gates, there exists a Hamiltonian $H_U \in \mathcal{H}_{\mathbf{c}} \otimes \mathcal{H}_{\mathbf{o}}$ acting on a clock and output register, i.e.*

$$H_U = \sum_{j=1}^N \sqrt{j(N-j+1)} (|j\rangle\langle j-1|_{\mathbf{c}} \otimes U_j + \text{H.C.}), \quad (\text{F1})$$

where H.C. means Hermitian conjugate terms. The quantum dynamic evolution governed by H_U and $t = \pi/2$ yields the original quantum circuit output as

$$e^{-iH_U t} |0\rangle_{\mathbf{c}}^{\otimes \log(N)} |\psi_0\rangle_{\mathbf{o}} = |N\rangle_{\mathbf{c}} \otimes U |\psi_0\rangle_{\mathbf{o}}. \quad (\text{F2})$$

With these ingredients (Lemma F.1 and Lemma F.3), we are ready to prove a lower bound on Qcc_2 .

Theorem F.1 (Lower bound on Qcc_2 of d-QS, Theorem 4 in maintext). *For any positive integer $\tau \leq n$, there exists a sparse Hamiltonian H acting on $2n + \log(n) + 1$ qubits with $\|H\| = \tau$ and a constant evolution time $t = \pi/2$ such that the (bounded-error) quantum communication complexity (Qcc_2) of any generic oblivious Γ -partite protocol for distributed quantum simulation (d-QS) scales at least linearly in scaled evolution time, that is, $\text{Qcc}_2(\text{d-QS}) = \Omega(\Gamma \|H\| t)$.*

Proof of Theorem F.1. Given an instance of $\text{IP}_n(\mathbf{y}, \mathbf{z})$, one can write a quantum circuit composed of n Toffoli gates to evaluate $\text{IP}(\mathbf{y}, \mathbf{z})$ by two parties (Alice and Bob) as in the L.H.S of Fig. 10. By Lemma F.3, given a quantum circuit, a Hamiltonian H_{IP} can be constructed as

$$H_{\text{IP}} := \sum_{s \in [n]} \sqrt{s(n-s+1)} |s\rangle\langle s-1|_{\mathbf{c}} \otimes \text{Toffoli}(y_s, z_s) + \text{H.C.} \quad (\text{F3})$$

of which the dynamics with the scaled evolution time $\|H_{\text{IP}}\| t = n\pi/2$ perfectly yields the circuit output. According to the known lower bound of quantum communication complexity for IP_n is $\Omega(n)$ (Lemma F.1), any distributed quantum simulation algorithm needs at least $\Omega(n)$ communication. For any positive integer $\tau < n$, we can similarly map an instance $\text{IP}_\tau(\mathbf{y}, \mathbf{z})$ to a Hamiltonian with $\|H_{\text{IP}}\| = \tau$ such that the linear lower bound also applies. It is worth noting

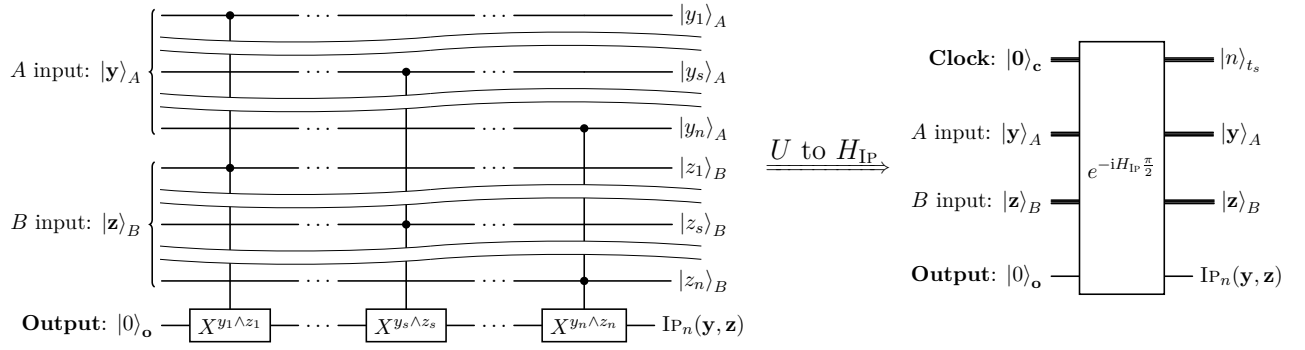


FIG. 10. The construction from an $\text{IP}_n(\mathbf{y}, \mathbf{z})$ circuit (instance) to a Hamiltonian evolution (instance) with H_{IP} and initial state $|0\rangle_c \otimes |\mathbf{y}\rangle_A \otimes |\mathbf{z}\rangle_B \otimes |0\rangle_o$. Alice (Bob) encodes an n -bit string $\mathbf{y}$ ($\mathbf{z}$) with n -qubit register and the output register consists of one qubit to store the inner product of two bit-strings. The L.H.S circuit consists of n Toffoli gate controlled by every pair of qubits of Alice and Bob, that is, apply an X gate to the output register when both y_s and z_s are 1. By the circuit-to-Hamiltonian map, a Hamiltonian H_{IP} can be constructed by introducing $\log(n)$ -qubit ancilla qubits as a clock register such that the dynamics of H_{IP} with $t = \pi/2$ yields the inner product of Alice and Bob's input bit-strings.

that in the proof of No fast-forwarding theorem, the input $\mathbf{x}$ is encoded into the Hamiltonian $H_{\mathbf{x}}$ and the evolution is simulated by querying entries of $H_{\mathbf{x}}$. In contrast, in this proof, the input $\mathbf{y}$ ($\mathbf{z}$) is encoded into Alice (Bob)'s quantum state, and the Hamiltonian H_{IP} is public to two parties, which matches the realistic scenario. For Γ -partite case, by Lemma F.2, our lower bound can be directly extended to $\Omega(\Gamma\|H\|t)$ by replacing IP_n with $\text{IP}_{n,\Gamma}$ and replacing the Toffoli gates in Fig. 10 with multi-partite controlled-NOT gates. Here, we underscore that for the task of quantum simulation dynamics, the amount of communication is fixed independently of the input. Therefore, it fits within the oblivious protocol context and our lower bound on party number Γ is optimal. $\square$